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Functional Requirement Specifications (FRS) for the I.M.O Strategic Engagement Platform 2026

Submitted to:

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1 Background

1.1 The International Maritime Organisation- Structure and India's Engagement

The International Maritime Organization (IMO) is a specialized agency of the United Nations responsible for regulating international shipping, enhancing maritime safety, preventing marine pollution, and addressing emerging challenges including greenhouse gas emissions, digitalization, and autonomous vessels. As a founding Member State and an actively engaged maritime nation, India participates at the highest levels across all IMO bodies.

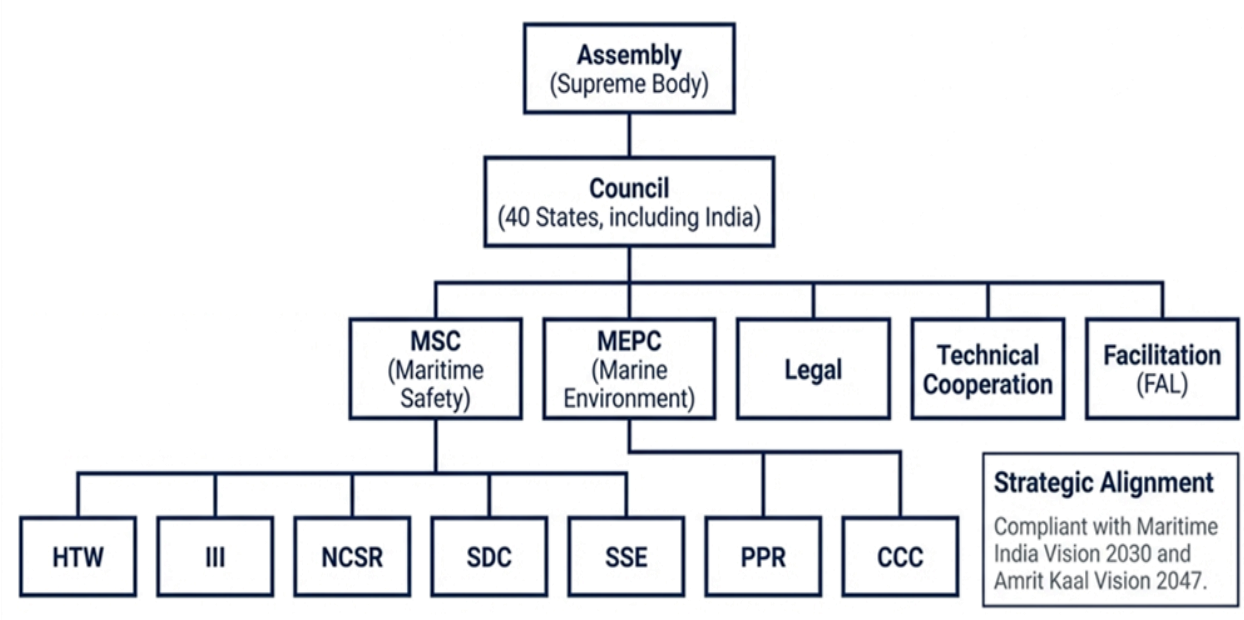
To carry out its wide-ranging mandate, the IMO is supported by a robust governance and technical structure, comprising an Assembly, a Council, and five principal technical committees, further assisted by specialized sub-committees.

The technical work of the International Maritime Organization is carried out by five principal Committees. These include:

a. The Maritime Safety Committee (MSC)
b. The Marine Environment Protection Committee (MEPC)
c. The Legal Committee
d. The Technical Cooperation Committee, for capacity building
e. The Facilitation Committee, to simplify the documentation and formalities required in international shipping. The Committees meet once or twice a year attended by Member States and NGOs

The MSC and MEPC are assisted in their work by several sub-committees which are open to all Member States. The committees are:

a. Sub-Committee on Human Element, Training and Watchkeeping (HTW)
b. Sub-Committee on Implementation of IMO Instruments (III)
c. Sub-Committee on Navigation, Communications and Search and Rescue (NCSR)
d. Sub-Committee on Pollution Prevention and Response (PPR)
e. Sub-Committee on Ship Design and Construction (SDC)
f. Sub-Committee on Ship Systems and Equipment (SSE)



Note:

- HTW: Human Element, Training and Watchkeeping
- III: Implementation of IMO Instruments
- NCSR: Navigation, Communications and Search & Rescue
- PPR: Pollution Prevention and Response
- SDC: Ship Design and Construction
- SSE: Ship Systems and Equipment
- CCC: Carriage of Cargoes and Containers

India actively participates in the International Maritime Organization (IMO), which governs global standards for maritime safety, security, and environmental sustainability. India's engagement is managed through the International Cooperation (IC) Branch of the Directorate General of Shipping (DGS), Ministry of Ports, Shipping & Waterways (MoPSW). The IC Branch coordinates inter-ministerial consultations, drafts India's positions on agenda items, manages delegations, and ensures timely, high-quality submissions to IMO and allied international bodies including ILO (International Labour Organization) and IMSO (International Maritime Satellite Organization).

To strengthen its engagement, **India has established Shadow Committees that mirror IMO structures, enabling coordinated preparation, research, and stakeholder consultation. These committees support the official delegation by providing inputs, drafting papers, and assisting during IMO sessions.**

Existing coordination mechanisms rely heavily on manual processes and dispersed communication channels, leading to inefficiencies, delays, and lack of traceability. To overcome these limitations, the Directorate General of Shipping **proposes the development of a unified digital engagement platform that streamlines workflow, enhances collaboration, and improves overall effectiveness**.

In addition to supporting end-to-end committee management, the proposed platform is envisaged as a comprehensive knowledge and governance platform for managing India's strategic engagement with international maritime organizations. The solution shall facilitate not only operational coordination but also

institutional knowledge preservation, regulatory tracking, document governance, stakeholder collaboration, and data-driven decision-making. The platform shall support the complete lifecycle of IMO, ILO, IMSO, Joint Working Groups (JWGs), bilateral engagements, and future international cooperation initiatives through configurable workflows and modular system architecture.

2 Existing Processes and Challenges

Based on internal and external stakeholder interactions, and our experience with government technology implementations, the following key internal challenges have been identified that the IMO Strategic Engagement Platform must address:

a. **Fragmented, manual document management**: Agenda papers, working documents, draft positions, and submission records are currently managed across email threads, individual drives, and paper files -with no centralized version control, audit trail, or search capability.

b. **Coordination gaps across multi-stakeholder workflows**: Inter-ministerial consultations (DGS, MMDs, MoPSW, MEA, MoEFCC) lack a structured digital channel, leading to last-minute position finalization, version conflicts, and inconsistent representations at IMO sessions.

c. **Knowledge erosion due to personnel transitions**: Institutional memory on India's historical positions, past submissions, committee deliberations, and country statements is not systematically captured or searchable, creating significant continuity risk.

d. **Meeting preparation inefficiency**: The end-to-end lifecycle from agenda distribution, review, feedback, consolidation, approval and submission lacks automation, resulting in manual effort, delays, and missed deadlines.

e. **Task tracking and accountability gaps**: Without a centralized task management and escalation system, pending actions from previous IMO sessions are difficult to track, leading to potential gaps in India's follow-through.

f. **Absence of real-time collaboration tools**: During live IMO committee sessions, DGS delegates lack a secure, real-time collaboration interface for last-minute position updates, interventions, and information sharing.

3 Purpose/ Objectives of the Assignment and Scope of Work

The proposed IMO Strategic Engagement Portal is envisaged as a comprehensive digital solution to strengthen and streamline the management of India's international maritime engagements. The key objectives of the assignment are as follows:

a. Develop a **unified platform** to manage **IMO committees, correspondence groups, agendas, interventions, and documentation**
b. Automate **end-to-end workflows**-from **agenda distribution and task assignment** to **drafting, feedback, approvals, and submission tracking**
c. **Real-time collaboration** among DGS officials and MoPSW stakeholders.
d. Build a **scalable, cloud-native solution** compliant with MeitY guidelines, extendable to other maritime bodies (e.g., ILO, IMSO)
e. Establish **role-based access, audit trails**, and monitoring to enhance accountability
f. Create a **secure, searchable repository** for IMO documents, historical interventions, and meeting outcomes
g. Provide **dashboards and reporting** for data-driven decision-making, bottleneck identification, and impact tracking.

3.1 Key Scope of Work

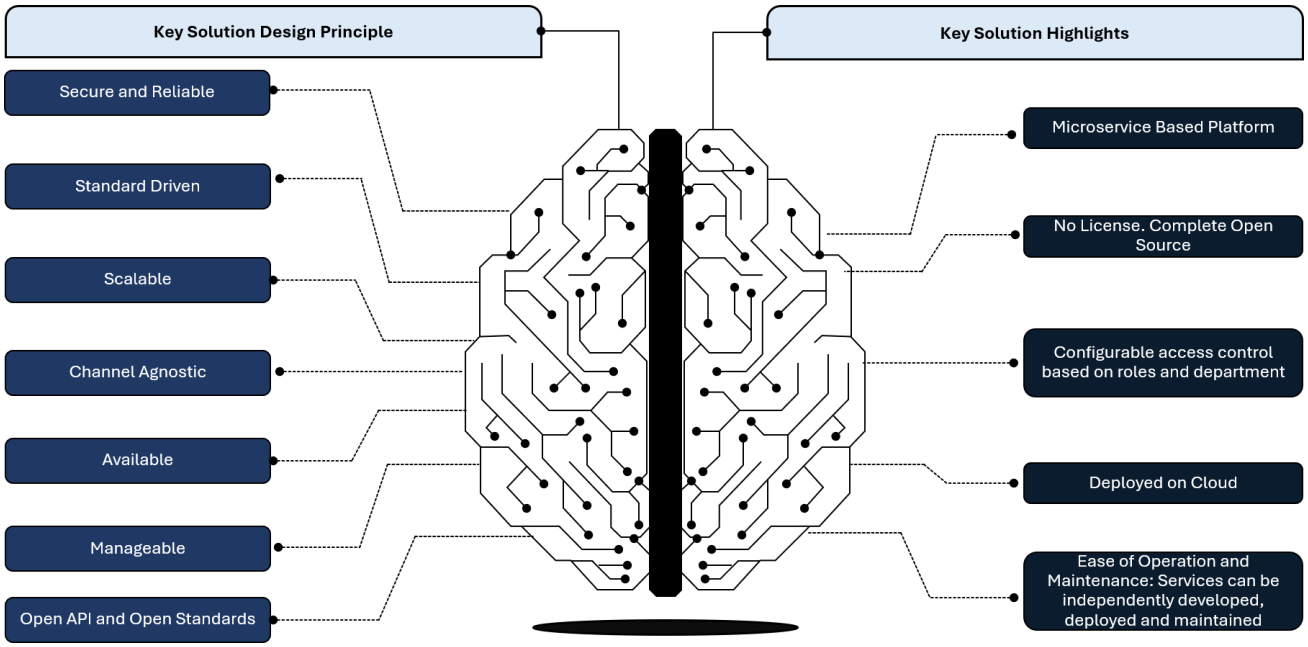
The key scope of work includes developing a secure, web-based collaborative portal for DGS with the mentioned features and functionalities:

1. Support interactions with global bodies such as the International Maritime Organization (IMO), the International Labour Organization (ILO), and the International Mobile Satellite Organization (IMSO).
2. Act as a centralized Document Management with a structured repository for circulars, agendas, meeting documents, position papers, and reports, ensuring easy access and version control
3. Enable multiple ministries and stakeholders to co-create, review, and finalize national submissions and positions in real time
4. Configurable approval workflows for streamlined clearance processes across departments, reducing turnaround times and manual dependencies.
5. Inbuilt tools to track upcoming IMO/ILO/IMSO sessions, assign responsibilities, manage delegations, and monitor timelines for submissions.
6. Unified interface to track progress, responsibilities, and action items across ministries, agencies, and subject experts.
7. Built-in messaging and notification systems to facilitate timely and confidential discussions among authorized users.
8. Historical records of past engagements, decisions, and submissions for institutional memory and continuity.
9. Data security and confidentiality by granting access based on user roles and organizational responsibilities.
10. Dashboards and reports to monitor participation, document status, compliance, and overall engagement performance.
11. Designed to integrate with existing government IT systems for seamless data exchange and interoperability (if required)

4 Solution Overview

4.1 Key Features of the Platform

The proposed solution combines modern software engineering practices with enterprise-grade architecture principles to deliver a secure, scalable, and future-ready platform. By leveraging a microservices-based architecture, open-source technologies, cloud-native deployment, configurable security controls, and open APIs, the platform ensures flexibility, interoperability, and ease of maintenance throughout its lifecycle. The modular design not only supports current operational requirements but also provides a strong foundation for future enhancements, integration with external systems, and the adoption of emerging technologies, thereby safeguarding the organization's long-term digital transformation objectives.



4.2 Proposed System Architecture

Technology components are the key factors in providing seamless user experience, efficient results, and sustainable performance. The overall solution will cover the following major factors:

- Seamless UI/UX
- Micro service/API based architecture
- Ease of integration or expansion
- Data and Document Security
- Performance and Reliability
- Logs and Reports
- Dashboard and Monitoring
- Seamless usage via Web browsers
- On-Premises hosting compatibility

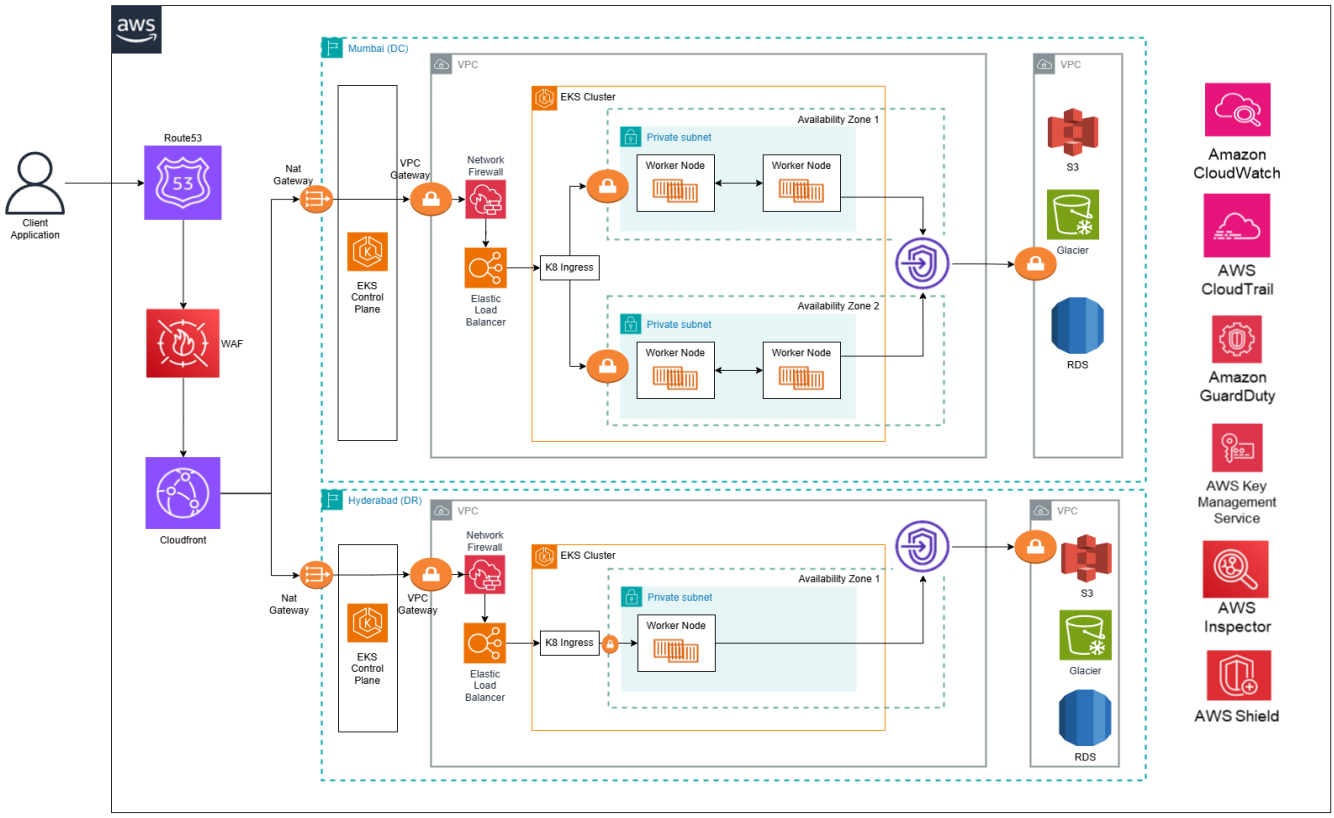
The proposed solution architecture encompasses a robust technology stack, seamless integration with key components, and a user-centric approach to meet the functional requirements of the solution along with MIS & Dashboard, Analytics, and Security requirements. The solution will be hosted on a MeitY empaneled cloud, leveraging state-of-the-art technologies to achieve scalability, performance, availability, and manageability.

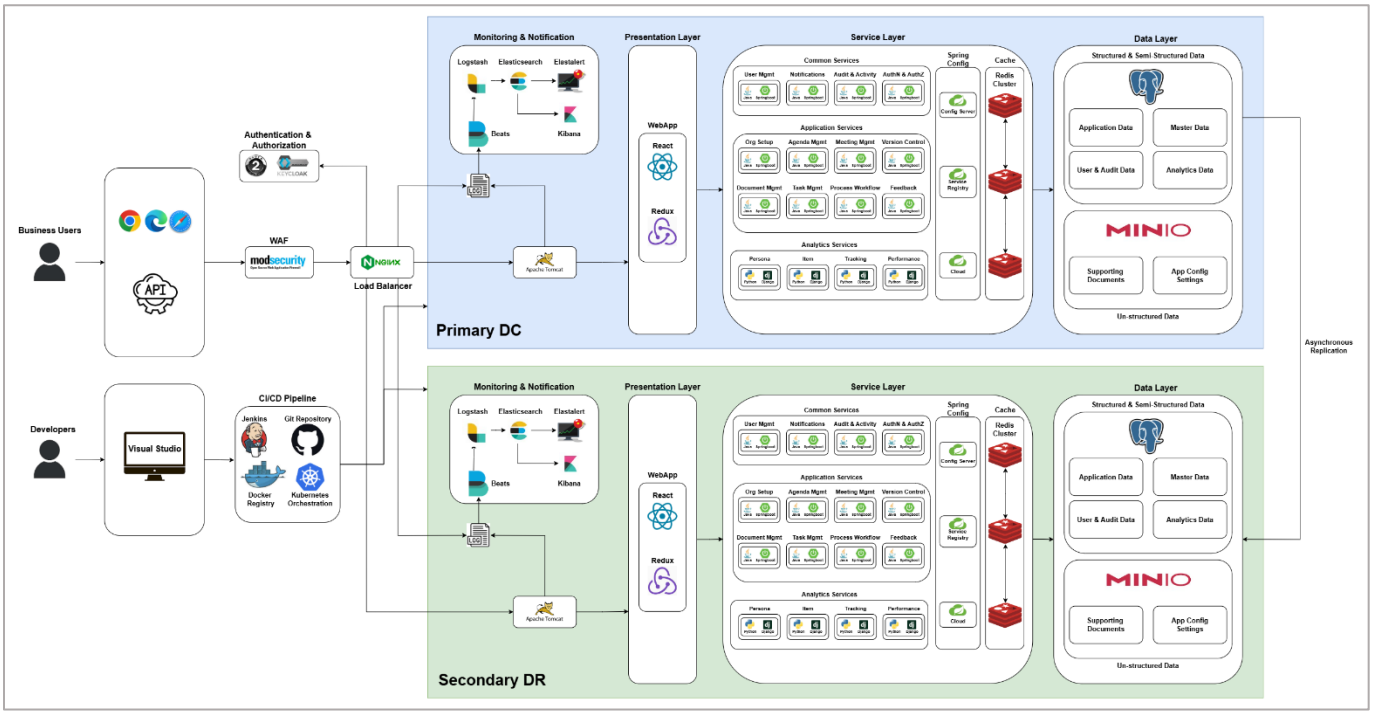
The diagram below describes the microservice based layered architecture of the solution which helps to design a robust futuristic application. The proposed Solution Architecture is a layered architecture based on Open Standards, and APIs with a clear separation of concerns. Each layer caters to a specific set of features and functionality with defined responsibilities and offers stated services to other layers. The proposed solution architecture encompasses a cutting-edge technology stack, seamless integration with key components, and a user-centric approach to meet the functional requirements.

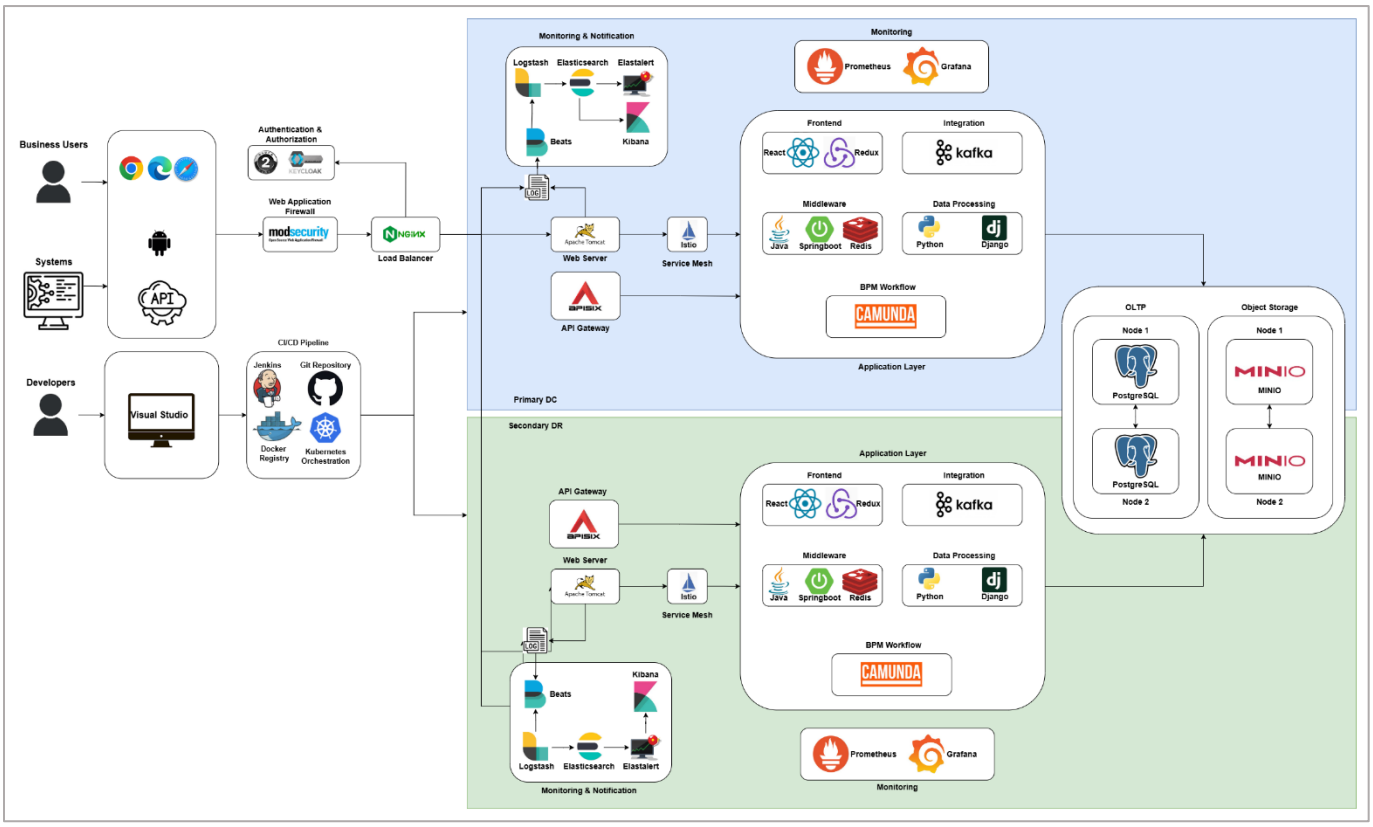
The overall technical architecture will be based on a centralized portal where all services can be accessed through single sign-on after authentication. The following diagram explains some of the key technologies, components, and infrastructure software used for the architecture of the solution. This is an indicative diagram based on current understanding. As the project progresses and additional requirements are identified through detailed analysis and stakeholder consultations, this will be reviewed, refined, and updated accordingly.

The proposed architecture has also been designed to support interoperability with existing and future Government digital platforms through secure APIs, standard data exchange mechanisms, and configurable integration services. The architecture shall support integration with Government identity providers (such as NIC SSO/ePramaan), document repositories, email gateways, notification services, and future maritime digital platforms, thereby ensuring long-term extensibility and interoperability without significant architectural modifications.

Solution Architecture Diagram

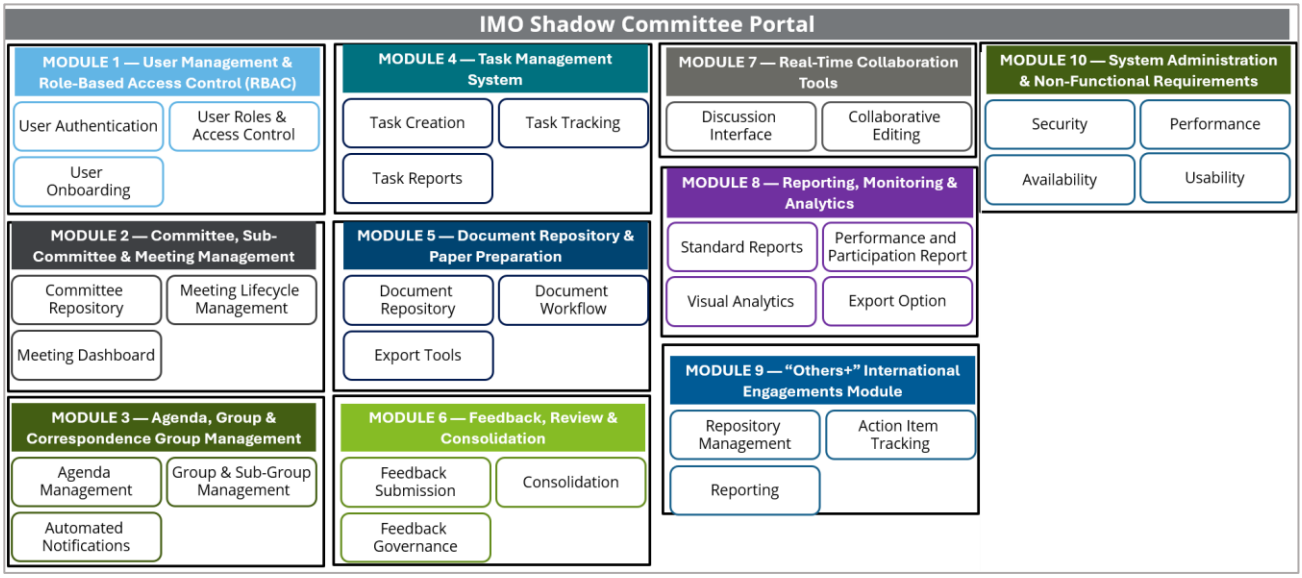






4.3 Core Functional Modules

The proposed IMO Shadow Portal will comprise ten integrated functional modules that will collectively support the end-to-end management of committees, correspondence groups, agenda items, documentation, stakeholder collaboration, reporting, and system administration. Each module is designed to address specific business requirements while seamlessly integrating with the overall platform.



a. Module 1: User Management & Role-Based Access Control (RBAC)

This module shall establish the foundation for secure access to the portal by managing user identities, authentication, and authorization. It enables user onboarding, role assignment, and configurable access permissions based on organizational hierarchy, departments, and responsibilities, ensuring that users can only access information relevant to their roles.

Key Features:

- User Authentication
- User Onboarding
- Role-Based Access Control (RBAC)

b. Module 2: Committee, Sub-Committee & Meeting Management

This module shall support the complete lifecycle management of IMO committees, sub-committees, and meetings. It maintains committee repositories, facilitates meeting scheduling and coordination, and provides dashboards for tracking meeting status, participation, and outcomes.

Key Features:

- Committee Repository
- Meeting Lifecycle Management
- Meeting Dashboard

c. Module 3: Agenda, Group & Correspondence Group Management

This module shall enable efficient management of agenda items, working groups, and correspondence groups. It facilitates the creation and tracking of agendas, manages group memberships, and automates notifications to ensure timely communication among stakeholders.

Key Features:

- Agenda Management
- Group & Sub-group Management
- Automated Notifications

d. Module 4: Task Management System

The Task Management module shall enable assignment, monitoring, and tracking of activities arising from committee discussions and decisions. It provides visibility into task ownership, progress, deadlines, and completion status, thereby improving accountability and timely execution.

Key Features:

- Task Creation
- Task Tracking
- Task Reports

e. Module 5: Document Repository & Paper Preparation

This module shall serve as the centralized repository for all committee-related documents, including submissions, working papers, information papers, reports, and reference materials. It also supports document workflows, version control, review processes, and document export functionalities.

Key Features:
- Centralized Document Repository
- Document Workflow Management
- Export Tools

f. Module 6: Feedback, Review & Consolidation

The module shall streamline the process of collecting comments and feedback from various stakeholders on agenda items, draft papers, and committee documents. It supports review workflows, governance mechanisms, and consolidation of inputs to facilitate the preparation of unified national positions.

Key Features:
- Feedback Submission
- Feedback Governance
- Consolidation of Comments

g. Module 7: Real-Time Collaboration Tools

This module shall promote effective collaboration among stakeholders by providing online discussion forums and collaborative editing capabilities. It enables multiple users to review, discuss, and refine documents in real time, reducing manual coordination and improving decision-making efficiency.

Key Features:
- Discussion Interface
- Collaborative Editing

h. Module 8: Reporting, Monitoring & Analytics

The Reporting and Analytics module shall provide comprehensive dashboards and analytical reports to support informed decision-making. It offers standard reports, performance monitoring, participation analysis, visual dashboards, and export capabilities for management reporting.

Key Features:
- Standard Reports
- Performance & Participation Reports
- Visual Analytics Dashboards
- Export Options

i. Module 9: International Engagements ("Others+") Module

This module shall extend the portal's capabilities beyond IMO committee management by supporting the tracking and management of other international maritime engagements, forums, and collaborative initiatives. It maintains repositories of relevant documents, tracks action items, and generates reports for effective monitoring.

Key Features:
- Repository Management
- Action Item Tracking
- Reporting

j. Module 10: System Administration & Non-Functional Requirements

This module shall provide administrative controls and system-wide configuration capabilities while ensuring compliance with non-functional requirements. It focuses on system security, performance optimization, availability, usability, monitoring, and overall platform administration to ensure reliable and efficient operations.

Key Focus Areas:

- Security Management
- Performance Optimization
- High Availability
- Usability and System Administration

4.4 Cross-Cutting Enterprise Capabilities

In addition to the functional modules described above, the platform shall provide several enterprise-wide capabilities that operate across all modules to improve governance, compliance, institutional memory, and operational efficiency.

a. Enterprise Search and Knowledge Discovery

The platform shall provide a centralized enterprise search capability enabling users to search across committees, meetings, agenda items, documents, interventions, correspondence groups, national positions, reports, action items, and archived records. Search functionality shall support metadata-based filtering, keyword search, OCR-based indexing of scanned documents, document classification, and advanced search operators.

b. Stakeholder Organization Registry

The platform shall maintain a centralized registry of participating organizations including Ministries, Government Departments, Classification Societies, Industry Associations, Research Institutions, Academic Organizations, Consultants, International Organizations, and other approved stakeholders. The registry shall maintain organization profiles, designated representatives, areas of expertise, committee participation history, and contact information to facilitate stakeholder management across committees.

c. Institutional Knowledge Repository

The platform shall establish a centralized institutional knowledge repository preserving committee history, historical interventions, national positions, working papers, correspondence group outputs, meeting reports, decisions, regulatory amendments, and lessons learnt. The repository shall facilitate knowledge reuse, historical reference, policy continuity, and improved preparation for future international engagements.

4.5 Security Framework

The proposed solution has been designed based on a comprehensive understanding of the infrastructure and security requirements of the engagement. The security framework aims to ensure the confidentiality, integrity, and availability of application data while providing a secure, resilient, and compliant operating environment. The application shall be hosted exclusively on a MeitY-empanelled Cloud Service Provider (CSP) in accordance with Government of India guidelines under the MeghRaj Coud Initiative. The deployment architecture shall comprise four logically segregated environments, namely Development, User Acceptance Testing (UAT), Production, and Disaster Recovery (DR), each maintained with independent access controls, security configurations, and administrative boundaries to ensure operational isolation and secure application lifecycle management.

The solution architecture shall be aligned with the prescribed Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirements. The platform shall be engineered to achieve these objectives through robust infrastructure planning, encrypted backup mechanisms, secure storage of backup data, and periodic Disaster Recovery (DR) drills conducted after Go-Live to validate recovery procedures and ensure business continuity in the event of system failures or disasters.

The platform shall implement information classification and access governance mechanisms to ensure that documents are handled according to their confidentiality and sensitivity. Documents may be categorized based on configurable classification levels such as Public, Internal, Confidential, Restricted, and Sensitive. Access permissions, document sharing, approval workflows, watermarking, download controls, and archival policies may be enforced based on document classification and user authorization levels.

The security framework shall comply with the applicable provisions of ISO/IEC 27001, ISO/IEC 27017, ISO/IEC 27018, CERT-In guidelines, and the Digital Personal Data Protection (DPDP) Act, 2023. Compliance shall be ensured through the implementation of appropriate security controls and governance mechanisms throughout the design, development, deployment, and operational phases of the project. As part of the security assurance process, the solution shall undergo three mandatory security audits at the Pre-Go-Live, Mid-Term, and Final stages of the project. Any critical or high-severity observations identified during these audits shall be addressed and resolved prior to acceptance of the security compliance certificate, thereby ensuring that the application meets the required security, compliance, and operational standards before production deployment and throughout the project lifecycle.

4.6 High-Level Workflow Overview

The workflow represents the high-level lifecycle of committee preparation, document development, approval, meeting execution, reporting, regulatory follow-up, and institutional knowledge preservation. Detailed workflows for each business process are described in Section 8.



5 User Roles and Access Management

The IMO Strategic Engagement Portal shall implement a comprehensive Role-Based Access Control (RBAC) framework to ensure that users are granted access only to the information and functionalities required for their assigned responsibilities. Each user shall be assigned one or more roles by the System Administrator, with permissions configured according to organizational hierarchy, committee responsibilities, and business processes. The RBAC framework will support secure access to committees, meetings, agenda items, documents, workflows, reports, and administrative functions while maintaining data confidentiality and operational integrity.

The platform shall support configurable role hierarchies, delegated authority, temporary role assignments, and user lifecycle management to ensure continuity of operations during officer transfers, leave, retirement, or organizational restructuring. Users may be assigned one or more roles, subject to authorization, and all role assignments shall remain fully auditable throughout their lifecycle.

5.1 Super User (DGS/DDG International Cooperation)

The Super User shall possess the highest level of administrative privileges within the portal and shall be responsible for overall governance, monitoring, and configuration of the system. This role shall oversee all committees, users, workflows, documents, reports, and system settings. The Super User shall have

authority to configure workflows, manage organizational roles, generate system-wide reports, monitor portal performance, and administer user access across all modules. Super users shall be enabled to configure and manage core governance and system administration parameters, including enterprise search settings, metadata standards, document classification schemes, document retention schedules, stakeholder registry, delegated authority framework, and regulatory master data. The system shall also provide institutional performance dashboards to facilitate monitoring of performance, operational metrics, and compliance indicators, thereby supporting effective governance and continuous performance improvement.

5.2 Administrator

The Administrator shall be responsible for the day-to-day administration of the portal and operational management of committee activities. Responsibilities shall include creation and management of committees, meetings, users, agenda items, documents, task assignments, notifications, and monitoring of workflow progress. The Administrator shall also manage document repositories, oversee task execution, and ensure timely completion of committee preparation activities.

The system shall enable the Administrator to perform comprehensive user and organizational administration functions, including user lifecycle management, account suspension and reactivation, membership renewal, and management of stakeholder organizations. The Administrator shall also be responsible for monitoring committee performance and member attendance, as well as creating, updating, and maintaining committee charters. These functionalities ensure effective governance, streamlined administrative operations, and efficient management of institutional and committee-related activities.

5.3 Delegation Leader

The Delegation Leader shall oversee the preparation and coordination of committee submissions for the assigned delegation. The role shall review consolidated feedback, monitor progress of agenda preparation, approve draft interventions and documents, manage delegation activities, and provide final recommendations prior to submission. The Delegation Leader shall also monitor meeting readiness and overall delegation performance through dedicated dashboards and reports.

The system shall enable the Delegation Leader to validate annual committee memberships, approve the engagement of external experts, and delegate approval authority to designated officials during periods of absence. The system shall also provide facilities to review committee and delegation readiness through dedicated dashboards and monitor the progress of intervention preparation to ensure timely coordination, effective participation, and informed decision-making for IMO engagements.

5.4 Coordinator

The coordinator shall manage the operational activities of assigned committees, working groups, and correspondence groups. Responsibilities shall include creation of sub-groups, assignment of agenda items, allocation of tasks to members, monitoring task completion, collecting stakeholder feedback, consolidating comments, and facilitating collaboration among committee participants. The coordinator shall ensure timely completion of assigned activities and support the Delegation Leader during document preparation.

In addition to coordinating committee activities, the Coordinator shall maintain dedicated correspondence group workspaces to facilitate collaboration among members, maintain submission trackers to monitor the progress of agenda-related documents and deliverables, monitor readiness indicators to assess the status of meeting preparation and pending activities, and track regulatory follow-up actions arising from

committee decisions to ensure timely implementation and closure of assigned action items.

5.5 Convener

The Convener shall act as the functional lead for a specific committee, working group, or correspondence group. The role shall coordinate discussions, review agenda-specific inputs, facilitate stakeholder engagement, validate technical comments, and ensure that submissions are aligned with committee objectives. The Convener shall collaborate closely with Coordinators and Delegation Leaders to finalize committee positions and recommendations.

The system shall enable the Convener to validate technical positions, maintain and update the issue register, and coordinate correspondence group activities. These functionalities shall support effective technical coordination, issue tracking, and collaboration to ensure timely development and review of committee outputs.

5.6 Member

Members shall participate in committee activities by reviewing assigned agenda items, accessing reference documents, preparing comments, participating in discussions, completing assigned tasks, and submitting feedback within specified timelines. Members shall have access only to committees, documents, and tasks assigned to them and shall be able to track the status of their submissions through personalized dashboards.

The system shall enable Members to participate in correspondence groups, submit interventions for review and consideration, and actively engage in discussion rooms to facilitate collaboration, knowledge sharing, and effective contribution to committee activities.

5.7 Consultant/Support Staff

Consultants and Support Staff shall provide technical, research, documentation, and administrative assistance to committee activities. Depending upon the permissions assigned, they may prepare draft documents, upload supporting material, assist in document management, participate in discussions, generate reports, and support coordinators in monitoring task progress. Access shall be limited to assigned committees and activities.

The system shall enable Consultants to participate in committee activities strictly based on the permissions assigned to them. Consultants shall not have approval authority or administrative rights and shall be limited to accessing and contributing only to the functions explicitly authorized for their role.

5.8 Ministry Reviewer

The Ministry Reviewer shall review agenda papers, interventions, policy documents, and consolidated feedback submitted for ministry-level examination. The role shall provide comments, recommendations, and observations on draft submissions before final approval. Ministry Reviewers shall primarily have review and comment privileges and shall not modify committee configurations or administrative settings.

The system shall enable the Ministry Reviewer to review national positions, assess regulatory implications, and evaluate policy consistency of proposed submissions and interventions. These functionalities shall support informed review and ensure alignment with applicable regulations, government policies, and strategic objectives.

5.9 Access Rights Matrix

Module	Super User	Administrator	Delegation Leader	Coordinator	Convener	Member	Consultant	Ministry Reviewer
Dashboard	Full	Full	Assigned	Assigned	Assigned	Personal	Assigned	Review
User Management	Full	Manage	View	View	No	No	No	No
Committee Management	Full	Full	View	Manage Assigned	View	View	View	View
Meeting Management	Full	Full	View	Manage	View	View	View	View
Agenda Management	Full	Full	Review	Manage	Manage	View	View	Review
Document Repository	Full	Full	Review	Upload/Manage	Upload	View	Upload	Review
Task Management	Full	Full	Monitor	Assign	Assign	Update Own	Update Assigned	View
Feedback Management	Full	Full	Consolidate	Consolidate	Review	Submit	Submit	Review
Approval Workflow	Full	Manage	Approve	Recommend	Recommend	No	No	Review
Reports & Analytics	Full	Full	Assigned	Assigned	Assigned	Personal	Assigned	Review
System Configuration	Full	Limited	No	No	No	No	No	No
Stakeholder Organization Registry	Full	Manage	View	View	View	No	View	View
Committee Charter Repository	Full	Manage	Review	Manage	View	View	View	Review
Committee Performance Dashboard	Full	Full	Monitor	Monitor	View	Personal	View	Review
Attendance & Participation Management	Full	Manage	Monitor	Manage	View	View Own	View	Review
Committee Output Repository	Full	Manage	Review	Upload/Manage	Upload	View	Upload	Review
Correspondence Workspace	Full	Manage	Monitor	Manage	Participate	Participate	Participate	View
Submission Tracker	Full	Manage	Monitor	Manage	View	View Assigned	View Assigned	Review
Regulatory Follow-up Tracking	Full	Manage	Monitor	Monitor	Update	Update Assigned	Update Assigned	Review
Readiness Dashboard	Full	Full	Full	Monitor	View	Personal	View	Review

5.10 Role Permission Matrix

Permission	Super User	Admin	Leader	Coordinator	Convener	Member	Consultant	Ministry Reviewer
Create	✓	✓	Limited	✓	Limited	✗	Limited	✗
View	✓	✓	✓	✓	✓	✓	✓	✓
Edit	✓	✓	Limited	✓	Limited	Own	Assigned	Comment Only
Delete	✓	✓	✗	Limited	✗	✗	✗	✗
Approve	✓	Limited	✓	Recommend	Recommend	✗	✗	Review
Assign Tasks	✓	✓	✓	✓	Limited	✗	✗	✗
Generate Reports	✓	✓	✓	Assigned	Assigned	Personal	Assigned	Review
Configure System	✓	Limited	✗	✗	✗	✗	✗	✗
Manage Committee Membership	✓	✓	Approve	Update	Recommend	✗	✗	✗
Approve External Experts	✓	Process	✓	Recommend	Recommend	✗	✗	✗
Delegate Approval Authority	✓	Configure	✓	Recommend	✗	✗	✗	✗
Manage Stakeholder Organizations	✓	✓	View	View	✗	✗	✗	✗
Manage Correspondence Workspace	✓	✓	Monitor	✓	Participate	Participate	Participate	✗
Monitor Readiness Dashboard	✓	✓	✓	✓	View	Personal	View	Review
Track Regulatory Follow-up	✓	✓	✓	✓	Update	Update Own	Update Assigned	Review
Manage Committee Charter	✓	✓	Approve	Edit	Review	View	View	Review

5.11 Delegation of Authority Matrix

The Delegation of Authority (DoA) defines the approval hierarchy and decision-making responsibilities across the portal.

Activity	Member	Convener	Coordinator	Delegation Leader	Administrator	Super User
Review Agenda	Prepare Comments	Validate	Consolidate	Approve	Monitor	Oversight
Prepare Documents	Draft	Review	Consolidate	Approve	Monitor	Oversight
Assign Tasks	✗	Limited	✓	✓	✓	✓

Create Meetings	✗	✗	Limited	✓	✓	✓
Create Committees	✗	✗	✗	✗	✓	✓
Manage Users	✗	✗	✗	✗	✓	✓
Configure Workflow	✗	✗	✗	✗	Limited	✓
Approve Final Submission	✗	Recommend	Recommend	✓	✓	✓
Generate Reports	Personal	Assigned	Assigned	Committee	Organization	System-wide
Validate Annual Committee Membership	✗	Recommend	Review	Approve	Monitor	Oversight
Approve External Experts	✗	Recommend	Review	Approve	Register	Oversight
Delegate Approval Authority	✗	✗	Recommend	Authorize	Configure	Oversight
Manage Stakeholder Organizations	✗	✗	✗	Review	Manage	Oversight
Maintain Correspondence Workspace	Participate	Review	Manage	Monitor	Oversight	Oversight
Maintain Submission Tracker	Update Assigned	Review	Manage	Monitor	Monitor	Oversight
Monitor Readiness Indicators	View Own	Review	Monitor	Approve	Monitor	Oversight
Track Regulatory Follow-up	Update Assigned	Review	Monitor	Approve	Monitor	Oversight
Manage Committee Charter	View	Recommend	Update	Approve	Manage	Oversight
Monitor Committee Performance	View Own	Review	Monitor	Review	Manage	Oversight
Monitor Attendance & Participation	View Own	Review	Monitor	Review	Manage	Oversight

6 Functional Requirements

6.1 User Management and Role-Based Access Control (RBAC) Module

Require-ment ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
UM-001	Login Management	All Users	The system shall allow registered users to log in using their registered username/email and password.	Only active users shall be authenticated. Invalid login attempts shall be logged.	Username, Password	User Dashboard	High
UM-002	Multi-Factor Authentication	Super User, Administrator, Delegation Leader	The system shall support Multi-Factor Authentication (MFA) for privileged users and shall provide configurable MFA policies for any user handling sensitive or confidential information.	MFA shall be configurable by the administrator and mandatory for designated roles.	OTP/Auth App	Secure Login	High
UM-003	Role-Based Access Control	Administrator	The system shall assign permissions based on predefined user roles.	Users shall access only authorized committees, meetings, documents, and tasks.	Role Assignment	Access Permissions	High
UM-004	Single Sign-On (SSO)	All Users	The system shall support Single Sign-On (SSO) with Government identity providers.	SSO integration shall be configurable without affecting local authentication.	SSO Credentials	User Session	Medium
UM-005	Audit Logs	Administrator	The system shall record all user activities including login, logout, edits, approvals, uploads, and downloads.	Audit logs shall be immutable and accessible only to authorized administrators.	User Activity	Audit Report	High
UM-006	User Lifecycle Management	Administrator	The system shall support complete user lifecycle management including user activation, suspension, deactivation, reactivation, transfer, and account expiry.	User status changes shall require authorized administrative action and shall be recorded in the audit log. Suspended or expired users shall not be permitted to access the portal.	User Profile, Status Change Request	Updated User Status	High

UM-007	Stakeholder Organization Registry	Administrator	The system shall maintain a centralized registry of stakeholder organizations including Ministries, Government Departments, Research Institutions, Industry Associations, Classification Societies, Academia, Consultants, and International Organizations.	Every external user shall be mapped to an approved stakeholder organization. The stakeholder registry shall be configurable.	Organization Details	Stakeholder Master Database	High
UM-008*)*	Delegated Authority / Acting Officer	Administrator, Delegation Leader	The system shall support temporary delegation of approval authority to designated Acting Officers during the absence of approving authorities.	Delegation shall be time-bound, role-based, and fully auditable. Automatic expiry shall occur upon completion of the delegation period.	Delegation Request	Temporary Role Assignment	High
UM-009	Annual Membership Validation	Administrator, Delegation Leader	The system shall support annual validation and renewal of committee memberships.	Members whose validation is pending or expired shall be flagged for review. Renewal history shall be maintained.	Membership Details	Updated Membership Status	High
UM-010	Configurable Multi-Factor Authentication	Administrator	The system shall allow Multi-Factor Authentication (MFA) to be enabled for any user or user group handling confidential or sensitive information.	MFA policies shall be configurable based on role, committee, document classification, or organizational policy.	MFA Policy	MFA Configuration	High
UM-011	Government Identity Integration	All Users	The system shall support integration with Government Identity Providers including NIC Single Sign-On (SSO) and ePramaan authentication services, wherever applicable.	Authentication mechanisms shall comply with applicable Government authentication standards.	Government Identity Credentials	Authenticated Session	High
UM-012	User Activity Analytics	Administrator, Super User	The system shall generate analytical reports on user activities including login trends, committee participation, document	Analytics shall be generated using audit logs without modifying operational data. Access	Audit Logs	User Activity Dashboard	Medium

			contributions, approval actions, and task completion.	shall be restricted to authorized users.			

6.2 Committee, Sub-Committee and Meeting Management Module

Require-ment ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
CM-001	Committee Repository	Administrator	The system shall maintain a centralized repository of IMO Committees, Sub-Committees, Assembly, Council, and future international forums.	Repository shall support future expansion without structural modification.	Committee Details	Committee Repository	High
CM-002	Committee Creation	Administrator	The system shall allow creation of committees with metadata including name, category, chairperson, and members.	Committee names shall be unique.	Committee Details	Committee Record	High
CM-003	Committee Membership	Administrator	The system shall assign users to committees with predefined roles.	One user may belong to multiple committees.	User & Committee	Membership Mapping	High
CM-004	Meeting Dashboard	All Assigned Users	The dashboard shall display agenda items, documents, deadlines, tasks, participation status, and meeting readiness indicators.	Information shall be filtered based on user role.	Meeting Selection	Dashboard View	High
CM-005	Committee Charter Repository	Administrator	The system shall maintain a repository of committee charters, Terms of Reference (ToR), mandates, objectives, and governance documents for each committee, sub-committee, working group, and correspondence group.	Each committee shall have a configurable charter linked to the committee repository with version history.	Charter Details	Committee Charter Repository	Medium
CM-006	Committee History Repository	Administrator	The system shall maintain the historical evolution of committees including membership changes, leadership assignments, committee restructuring, and meeting history.	Historical records shall be retained for audit, reporting, and institutional knowledge purposes.	Committee Updates	Committee History	High

CM-007	Committee Performance Monitoring	Administrator, Delegation Leader	The system shall monitor committee performance using configurable performance indicators including meeting preparedness, agenda completion, task completion, participation, and submission timelines.	Performance metrics shall be generated automatically based on operational data.	Committee Activities	Performance Dashboard	Medium
CM-008	Committee Output Repository	Administrator	The system shall maintain a centralized repository of committee outputs including national positions, interventions, working papers, information papers, reports, recommendations, and meeting outcomes.	Outputs shall be searchable and linked to the originating committee and meeting.	Approved Documents	Committee Output Repository	High
CM-009	Participant Attendance Management	Administrator, Coordinator	The system shall record and monitor attendance of committee members for meetings, working groups, correspondence groups, and related activities.	Attendance records shall be maintained for reporting and performance evaluation purposes.	Attendance Details	Attendance Register	Medium
CM-010	Meeting Readiness Dashboard	Delegation Leader, Coordinator	The system shall provide a Meeting Readiness Dashboard displaying completion status of agenda preparation, document drafting, feedback submission, approvals, pending tasks, and overall readiness indicators.	Readiness indicators shall be updated automatically based on workflow progress.	Meeting Data	Meeting Readiness Dashboard	High
CM-011	Historical Meeting Analytics	Super User, Administrator	The system shall provide analytical reports on historical meetings including participation trends, committee workload, agenda completion, document submissions, and committee performance over time.	Historical analytics shall be generated from archived meeting records and shall support configurable filtering.	Historical Meeting Data	Meeting Analytics Reports	Medium

6.3 Agenda, Working Group and Correspondence Group Management Module

Require-ment ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
AG-001	Agenda Creation	Coordinator	The system shall allow creation of agenda items linked to meetings, while capturing mandatory IMO metadata and supporting document associations	Every agenda item shall belong to only one meeting.	Agenda Details	Agenda Record	High
AG-002	Agenda Assignment	Coordinator	Agenda items shall be assigned to one or more responsible members.	Multiple assignees shall be supported.	Agenda, Members	Assignment Record	High
AG-003	Working Groups	Coordinator	Working Groups and Correspondence Groups shall be created against agenda items.	Each group shall have one Convener and one Coordinator.	Group Details	Working Group	Medium
AG-004	Agenda Metadata Management	Coordinator	The system shall maintain comprehensive metadata for every agenda item including Session Number, Agenda Number, Working Paper Number, Information Paper Number, Circular Letter Reference, Committee, Agenda Leader, Submission Deadline, National Position Status, Intervention Status, Regulatory Impact, and Follow-up Status.	Mandatory metadata fields shall be validated before an agenda item is published.	Agenda Metadata	Agenda Master Record	High
AG-005	IMO Document Linkage	Coordinator	The system shall allow agenda items to be linked with IMO Circular Letters, Working Papers, Information Papers, Draft Resolutions, and supporting reference documents.	One agenda item may be associated with multiple IMO documents.	IMO Documents	Linked Agenda Repository	High
AG-006	Submission Lifecycle Management	Coordinator, Delegation Leader	The system shall track the complete lifecycle of agenda-related submissions from draft preparation through review,	Submission status shall be updated automatically as	Submission Details	Submission Status Tracker	High

			approval, submission, revision, and final acceptance.	workflow stages are completed.			
AG-007	Correspondence Workspace	Coordinator	The system shall provide a dedicated collaborative workspace for each Correspondence Group to facilitate document preparation, discussions, draft reviews, and stakeholder collaboration.	Access shall be restricted to authorized Correspondence Group members.	Group Details	Correspondence Workspace	High
AG-008	Draft Circulation Register	Coordinator	The system shall maintain a register of all draft documents circulated for review within Working Groups and Correspondence Groups.	Every circulation shall record document version, recipients, circulation date, and response status.	Draft Documents	Draft Circulation Register	Medium
AG-009	Comment Consolidation Register	Coordinator	The system shall maintain a structured register of comments received during agenda preparation and correspondence group discussions.	All comments shall remain traceable to the contributor and document version.	Comments	Consolidated Comment Register	Medium
AG-010	Submission Tracker	Coordinator, Delegation Leader	The system shall monitor the progress of submissions against each agenda item, including pending drafts, approvals, submitted papers, and submission deadlines.	Dashboard indicators shall display submission status in real time.	Submission Data	Submission Dashboard	High
AG-011	Correspondence Timeline Monitoring	Coordinator	The system shall monitor milestones and timelines associated with Correspondence Group activities including draft circulation, review deadlines, comment submission, and final submission.	Automatic reminders and escalation shall be generated for overdue milestones.	Timeline Details	Timeline Dashboard	Medium
AG-012	Version Comparison Utility	Coordinator, Members	The system shall provide version comparison functionality to identify additions, deletions, and modifications between successive versions of agenda	Previous versions shall remain immutable and available for comparison.	Document Versions	Comparison Report	Medium

			documents and correspondence drafts.				
AG-013	Agenda Continuation Tracking	Coordinator	The system shall maintain continuity across successive committee sessions by linking current agenda items with previous discussions, historical interventions, decisions, submissions, and pending actions.	Historical references shall remain searchable and available for future meetings.	Historical Agenda Data	Agenda Continuity Repository	High

6.4 Task Management Module

Requirement ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
TM-001	Task Creation	Coordinator	Tasks shall be created against agenda items or documents.	Every task shall have an owner and due date.	Task Details	Task Record	High
TM-002	Reminder Engine	System	Automated reminders shall be sent before task due dates.	Reminder frequency shall be configurable.	Due Date	Notification	High
TM-003	Escalation	Coordinator	Overdue tasks shall be escalated to the next approval level.	Escalation rules shall be configurable.	Task Status	Escalation Alert	High
TM-004	SLA Monitoring	Administrator, Coordinator	The system shall monitor Service Level Agreement (SLA) timelines for all assigned tasks and track compliance against predefined completion targets.	SLA thresholds shall be configurable based on task category, priority, or committee requirements. Automatic alerts shall be generated for impending SLA breaches.	Task Details, SLA Parameters	SLA Compliance Dashboard	High
TM-005	Task Readiness Dashboard	Delegation Leader, Coordinator	The system shall provide a real-time dashboard displaying task completion status, pending tasks, overdue tasks, upcoming deadlines, and overall committee readiness.	Dashboard indicators shall be updated automatically based on task progress and workflow status.	Task Status	Task Readiness Dashboard	High
TM-006	Regulatory Follow-up Tasking	Coordinator	The system shall support creation and monitoring of regulatory follow-up tasks arising from IMO resolutions, amendments, committee decisions, and action items.	Regulatory tasks shall remain linked to the originating agenda item or meeting until formally closed.	Regulatory Action Details	Regulatory Task Register	High
TM-007	Task Aging Analysis	Administrator, Delegation Leader	The system shall monitor the aging of open tasks and classify them based on configurable age bands to support management	Aging calculations shall be performed automatically using the task creation date, due date, and current status.	Task Data	Task Aging Dashboard	Medium

			review and timely intervention.				

6.5 Document Repository and Paper Preparation Module

Requirement ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
DR-001	Document Upload	All Users	The system shall allow users to upload documents in supported formats.	File size and format restrictions shall be enforced.	Document File	Stored Document	High
DR-002	Version Control	All Users	The system shall maintain version history of documents.	Previous versions shall be retrievable and immutable.	Updated Document	Version History	High
DR-003	Document Classification	Coordinator	Documents shall be categorized based on committee, meeting, agenda, document type, confidentiality classification, and mandatory metadata prior to submission.	Classification shall be mandatory before submission.	Document Metadata	Classified Document	High
DR-004	Drafting Tool	Delegation Members	The system shall provide tools for drafting papers collaboratively.	Changes shall be tracked with user attribution.	Draft Content	Draft Document	Medium
DR-005	Approval Workflow	Administrator	Documents shall pass through configurable approval workflows before final submission.	Approval hierarchy shall be role-based.	Document	Approved Document	High
DR-006	Enterprise Search	All Users	The system shall provide enterprise-wide search capabilities to retrieve documents using keywords, metadata, document number, committee, meeting, agenda, author, document type, approval status, and other configurable filters.	Search results shall be filtered according to the user's access permissions under Role-Based Access Control (RBAC).	Search Criteria	Search Results	High
DR-007	OCR Search	All Users	The system shall support Optical Character Recognition (OCR) indexing of scanned PDF documents and images to	OCR indexing shall be performed automatically for supported file	Scanned Documents	Searchable Document Index	Medium

			enable full-text search across archived and uploaded documents.	formats after document upload.			
DR-008	Document Metadata Framework	Coordinator	The system shall maintain structured metadata for every document including Session Number, Committee, Agenda Number, Document Number, Document Type, Classification, Author, Version, Approval Status, Reviewer, Keywords, Creation Date, and Last Modified Date.	Mandatory metadata fields shall be validated before document publication or approval.	Document Metadata	Metadata Repository	High
DR-009	Watermarking	Administrator	The system shall support configurable watermarking of confidential, draft, review, and approved documents during viewing, download, and printing.	Watermarks shall be configurable based on document classification and approval status.	Watermark Policy	Watermarked Document	Medium
DR-010	Document Type Management	Administrator	The system shall support standardized document categories including Working Papers, Information Papers, Circular Letters, Draft Resolutions, Speaking Notes, Interventions, Briefing Notes, National Position Papers, Outcome Reports, and other configurable document types.	Document type selection shall be mandatory during document creation or upload.	Document Details	Classified Document	High
DR-011	Confidentiality Classification	Administrator	The system shall support configurable document classification levels such as Public, Internal, Confidential, Restricted, and Secret for controlling document access and handling.	Document access permissions shall be automatically enforced based on the assigned classification level and user role.	Classification Details	Classified Document	High
DR-012	Document Retention & Archival	Administrator	The system shall support configurable document retention schedules and archival policies in accordance with Government	Documents shall be retained, archived, or disposed of only in accordance with	Retention Rules	Archived Document	Medium

			record management requirements.	approved retention policies.			
DR-013	Document Relationship Management	Coordinator	The system shall support linking related documents including agenda papers, circular letters, interventions, national positions, working papers, correspondence, and meeting reports to provide complete document traceability.	One document may be linked to multiple meetings, agenda items, or related documents.	Linked Document Details	Document Relationship Map	Medium
DR-014	Document Preview & Online Viewer	All Users	The system shall provide an integrated document viewer enabling users to preview supported document formats without downloading them.	Viewing permissions shall comply with document classification and RBAC policies.	Document Selection	Online Document Viewer	Medium

6.6 Feedback, Review and Consolidation

Requirement ID	Module	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
FR-001	Feedback & Review	Feedback Capture	All Users	The system shall allow users to provide feedback on documents and agenda items.	Feedback shall be linked to specific items.	Feedback Content	Feedback Record	High
FR-002	Feedback & Review	Review Workflow	Coordinator	Feedback shall be reviewed and categorized by coordinators.	Only authorized users can approve or reject feedback.	Feedback	Review Status	High
FR-003	Feedback & Review	Consolidation	Delegation Leader	The system shall consolidate feedback into structured reports.	Duplicate feedback shall be merged automatically.	Feedback Data	Consolidated View	Medium
FR-004	Feedback & Review	Comment Disposition	Coordinator	The system shall support acceptance, rejection, or partial acceptance of individual comments	Every comment disposition shall be recorded with the corresponding	Feedback Comments	Comment Disposition Register	High

				received during the review process.	reviewer and status.			
FR-005	Feedback & Review	Justification Management	Coordinator, Delegation Leader	The system shall capture justification for comments that are rejected, modified, or partially accepted during the review and consolidation process.	Justification shall be mandatory where comments are rejected or significantly modified.	Reviewer Comments	Justification Repository	High
FR-006	Feedback & Review	Dissent Management	Delegation Leader	The system shall support recording and management of dissenting opinions where consensus is not achieved during committee deliberations.	Dissent notes shall remain linked to the relevant agenda item and document for future reference.	Dissent Comments	Dissent Register	Medium
FR-007	Feedback & Review	Historical Feedback Repository	All Users	The system shall maintain a searchable repository of historical feedback associated with meetings, agenda items, committees, documents, and contributors.	Access to historical feedback shall be governed by Role-Based Access Control (RBAC).	Search Criteria	Historical Feedback Records	High
FR-008	Feedback & Review	Stakeholder Analytics	Administrator, Delegation Leader	The system shall generate analytical reports on stakeholder participation, feedback contribution, review completion, and acceptance rates.	Analytics shall be generated automatically from submitted feedback and review records.	Feedback Records	Stakeholder Analytics Dashboard	Medium

6.7 Collaboration Module

Requirement ID	Module	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
CL-001	Collaboration	Messaging	All Users	The system shall provide messaging capabilities among users.	Messages shall be stored securely.	Message Content	Message Thread	Medium
CL-002	Collaboration	Discussion Forum	All Users	Users shall participate in discussion forums linked to agenda items.	Discussions shall be moderated by coordinators.	Discussion Post	Discussion Thread	Medium
CL-003	Collaboration	Notifications	System	The system shall send notifications for updates, tasks, and deadlines.	Notification preferences shall be configurable.	Event Trigger	Notification	High
CL-004	Collaboration	Agenda-wise Discussion Rooms	All Assigned Users	The system shall provide dedicated discussion rooms for each agenda item to facilitate focused technical discussions among assigned members.	Access shall be restricted to members assigned to the corresponding agenda item or working group.	Agenda ID	Agenda Discussion Workspace	High
CL-005	Collaboration	Real-Time Collaborative Editing	All Assigned Users	The system shall allow multiple authorized users to simultaneously edit draft documents and intervention papers in real time.	Concurrent editing shall preserve version history and prevent data conflicts.	Draft Document	Updated Collaborative Document	High
CL-006	Collaboration	Comments & Annotations	All Assigned Users	The system shall allow users to add comments, annotations, and review notes directly within documents and agenda items.	Comments shall be attributed to individual users and remain traceable throughout the review process.	Comment	Annotated Document	High
CL-007	Collaboration	Track Changes	All Assigned Users	The system shall automatically track additions, deletions,	All document changes shall be attributable to the	Document Changes	Change History	High

				modifications, and reviewer comments during collaborative document preparation.	user making the modification and preserved within version history.			
CL-008	Collaboration	Mentions & Alerts	All Users	The system shall support @mentions to notify specific users or groups during discussions and document reviews.	Mentioned users shall receive immediate notifications based on their notification preferences.	Mention Event	User Notification	Medium
CL-009	Collaboration	Presence & Activity Status	All Assigned Users	The system shall display the availability and collaboration status of users currently participating in document editing or discussion sessions.	Presence indicators shall be visible only to authorized participants.	User Session	Presence Status	Low

6.8 Reporting, Monitoring and Analytics Module

Requirement ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
RA-001	Standard Reports	Administrator	The system shall generate predefined reports on meetings, tasks, and documents.	Reports shall be exportable in multiple formats.	Report Parameters	Report Output	High
RA-002	Custom Reports	Administrator	Users shall create custom reports based on filters and criteria.	Access to data shall follow RBAC rules.	Filters	Custom Report	Medium
RA-003	Dashboard Metrics	All Users	The system shall display real-time metrics on dashboards.	Metrics shall refresh periodically.	System Data	Dashboard Widgets	High
RA-004	Agenda Readiness Dashboard	Delegation Leader, Coordinator	The system shall provide dashboard views indicating the preparation status of agenda items including document readiness, task completion, approvals, and pending activities.	Readiness indicators shall be automatically updated based on workflow progress.	Agenda Status	Agenda Readiness Dashboard	High

RA-005	National Position Dashboard	Delegation Leader, Ministry Reviewer	The system shall provide dashboard visibility of the preparation, review, approval, and submission status of National Positions for each agenda item.	Dashboard shall display current workflow stage and pending approvals.	National Position Data	National Position Dashboard	High
RA-006	Attendance & Participation Analytics	Administrator, Delegation Leader	The system shall generate attendance and participation reports for meetings, committees, and working groups.	Attendance statistics shall be automatically calculated from meeting participation records.	Attendance Records	Attendance Analytics	Medium
RA-007	Member Performance Dashboard	Administrator, Delegation Leader	The system shall monitor member performance based on agenda ownership, task completion, feedback submissions, review participation, interventions, and meeting attendance.	Performance indicators shall be configurable and generated automatically.	Member Activity Data	Performance Dashboard	High
RA-008	Approval Analytics	Administrator, Super User	The system shall generate reports on approval workflow performance, including approval timelines, pending approvals, bottlenecks, and approval delays.	Workflow analytics shall be generated from approval history and audit logs.	Approval Records	Approval Analytics	Medium
RA-009	Committee Performance Dashboard	Super User, Administrator	The system shall provide analytical dashboards to evaluate committee performance based on meeting readiness, agenda completion, submissions, interventions, and overall committee activities.	Performance metrics shall be configurable and periodically refreshed.	Committee Activity Data	Committee Dashboard	Medium
RA-010	Regulatory & Follow-up Monitoring	Delegation Leader, Coordinator	The system shall generate reports tracking regulatory follow-up actions, implementation status, pending amendments, and post-meeting action items.	Reports shall consolidate action status across committees and meetings.	Regulatory Action Data	Regulatory Monitoring Report	High

Key Performance Indicators (KPIs)

The Reporting and Analytics Module shall provide configurable Key Performance Indicators (KPIs) to support decision-making by senior management. Indicative KPIs shall include, but not be limited to:

- Meeting Attendance Percentage
- Agenda Ownership Status
- Task Completion Rate
- Feedback Contribution by Member
- Review Participation Statistics
- Approval Turnaround Time
- Approval Delay Analysis
- Submission Status
- Document Preparation Status
- Outstanding Action Items
- Historical Trend Analysis

6.9 Others+ International Engagement Module

Requirement ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
OT-001	Configuration	Administrator	The system shall allow configuration of system parameters.	Changes shall be logged and versioned.	Configuration Data	Updated Settings	High
OT-002	Integration	Administrator	The system shall integrate with external systems and APIs.	Integration shall follow secure protocols.	API Data	Integrated Output	High
OT-003	Backup & Recovery	Administrator	The system shall support automated backup and recovery mechanisms.	Backup schedules shall be configurable.	System Data	Backup Files	High
OT-004	International Action Tracking	Coordinator	The system shall track action items, commitments, resolutions, and follow-up activities arising from international meetings.	Every action item shall have an owner, target date, and status.	Action Details	Action Tracker	High
OT-005	ILO Workflow Management	Delegation Leader	The system shall support structured workflows for	Workflow stages shall be	ILO Meeting Details	ILO Workflow	High

			International Labour Organization (ILO) engagements including Special Tripartite Committee (STC), Joint Maritime Commission (JMC), Maritime Labour Convention (MLC) amendments, and wage negotiation activities.	configurable for different ILO processes.			
OT-006	IMSO Workflow Management	Delegation Leader	The system shall support structured workflows for International Mobile Satellite Organization (IMSO) engagements including Assembly meetings, LRIT activities, Advisory Committee meetings, and related submissions.	Workflow shall maintain complete meeting history and document traceability.	IMSO Meeting Data	IMSO Workflow	High
OT-007	Joint Working Group (JWG) Management	Coordinator	The system shall support creation and management of Joint Working Groups, bilateral cooperation initiatives, and regional maritime collaboration forums.	Each JWG shall maintain its own members, documents, tasks, and action register.	JWG Details	JWG Workspace	Medium
OT-008	Commitment & Resolution Monitoring	Delegation Leader	The system shall monitor implementation of international commitments, adopted resolutions, and agreed actions arising from international engagements.	Progress shall be periodically updated until formal closure.	Resolution Details	Commitment Tracker	High
OT-009	Outcome Reporting	Administrator	The system shall generate reports summarizing meetings, commitments, action items, agreements, and implementation status for international engagements.	Reports shall be exportable in PDF, Excel, and XML formats.	Reporting Parameters	Outcome Reports	Medium
OT-004	International Action Tracking	Coordinator	The system shall track action items, commitments, resolutions, and follow-up activities arising from international meetings.	Every action item shall have an owner, target date, and status.	Action Details	Action Tracker	High

Additional Features

The module shall additionally support:

ILO Features

- Special Tripartite Committee (STC) workflow management.
- Joint Maritime Commission (JMC) workflow management.
- Maritime Labour Convention (MLC) Amendment Tracking.
- Wage Negotiation Reference Repository.
- Historical ILO submissions and meeting repository.

IMSO Features

- IMSO Assembly management.
- Long Range Identification and Tracking (LRIT) activity tracking.
- Advisory Committee meeting management.
- IMSO document repository.
- Historical IMSO decision repository

Other International Engagements

- Bilateral maritime cooperation.
- Regional cooperation forums.
- Joint Working Groups (JWGs).
- International agreements and Memoranda of Understanding (MoUs).
- International commitment monitoring.
- Outcome monitoring and implementation tracking.

6.10 System Administration and Non-Functional Requirements

Requirement ID	Sub-Module	User Role	Functional Requirement	Business Rules / Logic	Inputs	Outputs	Priority
SA-001	Performance	System	The system shall support concurrent users without performance degradation.	Response time shall meet defined SLA thresholds.	User Load	Performance Logs	High
SA-002	Security	System	The system shall ensure data encryption and secure communication.	Encryption standards shall comply with regulations.	Data	Secure Data	High

SA-003	Availability	System	The system shall ensure high availability and uptime.	Downtime shall be minimized and monitored.	System Status	Availability Logs	High
SA-004	System Configuration	Administrator	The system shall allow configuration of organizational parameters, workflows, notification rules, approval hierarchies, and master data.	Configuration changes shall be logged, version controlled, and auditable.	Configuration Data	Updated System Configuration	High
SA-005	Integration Management	Administrator	The system shall support secure integration with Government platforms, identity providers, and external applications through standard APIs.	All integrations shall comply with secure API protocols and Government interoperability standards.	API Requests	Integrated Services	High
SA-006	Backup & Recovery	Administrator	The system shall support automated backup, restoration, and disaster recovery mechanisms.	Backup schedules, retention periods, and recovery procedures shall be configurable.	System Data	Backup Repository	High
SA-007	Government Identity Integration	Administrator	The system shall support integration with Government Single Sign-On services such as NIC SSO/ePramaan.	Authentication mechanisms shall comply with Government identity management standards.	Identity Credentials	Authenticated User Session	High
SA-008	Data Classification	Administrator	The system shall classify information based on confidentiality levels such as Public, Internal, Confidential, and Restricted.	Access permissions shall be automatically enforced based on document classification.	Document Metadata	Classified Information	High
SA-009	Record Retention Management	Administrator	The system shall implement configurable retention schedules for documents, audit logs, meeting records, and historical information.	Records shall be retained and disposed of in accordance with applicable Government record management policies.	Retention Policies	Archived Records	High
SA-010	Audit & Log Retention	Administrator	The system shall maintain audit logs, security logs, and operational logs for	Logs shall be tamper-resistant and accessible only to	System Logs	Audit Repository	High

			configurable retention periods.	authorized administrators.			
SA-011	Security Monitoring	Administrator	The system shall support continuous monitoring of application, infrastructure, and security events.	Alerts shall be generated for abnormal activities, security incidents, and operational failures.	Monitoring Events	Security Alerts	High
SA-012	Compliance Management	Administrator	The system shall support compliance with MeitY Guidelines, CERT-In Directions, GIGW 3.0, DPDP Act, ISO/IEC 27001, ISO/IEC 27017, and ISO/IEC 27018.	Compliance requirements shall be configurable and periodically reviewed.	Compliance Policies	Compliance Reports	High

The System Administration Module shall additionally support:

- Master data management for committees, organizations, ministries, user roles, and reference data.
- Workflow configuration and approval hierarchy management.
- Notification and reminder configuration.
- System health monitoring and diagnostics.
- Performance monitoring and capacity utilization reports.
- Scheduled backup and restoration management.
- API management and integration monitoring.
- Security policy configuration.
- Audit log monitoring and archival.
- Configuration version management.
- Compliance reporting and governance dashboards

7 Non- Functional Requirements

The IMO Strategic Engagement Portal shall be designed to meet the non-functional requirements outlined in this section to ensure a secure, reliable, scalable, high-performing, and user-friendly platform. These requirements define the quality attributes of the solution and establish measurable criteria for system performance, security, usability, interoperability, availability, and compliance.

7.1 Security Requirements

Requirement ID	Category	Requirement	Acceptance Criteria	Priority
NFR-SEC-001	Authentication	The system shall provide secure authentication using official email ID or approved Government identity credentials.	Only authenticated users shall be able to access the portal.	High
NFR-SEC-002	Authorization	The system shall implement Role-Based Access Control (RBAC) to restrict access to committees, meetings, documents, reports, and administrative functions based on assigned user roles.	Users shall access only authorized resources.	High
NFR-SEC-003	Encryption	All communication between client devices and the applications shall be encrypted using HTTPS with TLS 1.3 or higher.	All network traffic shall be transmitted over secure channels.	High
NFR-SEC-004	Audit Logging	The system shall maintain a complete audit trail of user activities including login, logout, uploads, downloads, edits, approvals, workflow actions, and feedback submissions.	Audit logs shall be tamper-resistant and accessible only to authorized administrators.	High
NFR-SEC-005	Security Compliance	The solution shall comply with applicable MeitY Cloud Security Guidelines and undergo periodic security assessments and vulnerability testing.	Security assessment reports shall demonstrate compliance.	High
NFR-SEC-006	Multi-Factor Authentication	The system shall support configurable Multi-Factor Authentication (MFA) for all users handling confidential or classified information, with mandatory enforcement for privileged roles.	MFA policies shall be configurable by the Administrator and enforced based on user role or document sensitivity.	High
NFR-SEC-007	Government Single Sign-On	The system shall support integration with Government identity providers including NIC Single Sign-On (SSO) / ePramaan, while retaining secure local authentication as a fallback mechanism.	Users shall be authenticated through approved Government identity services wherever applicable.	High
NFR-SEC-008	Data Classification	The system shall support classification of information assets and documents based on	Document access, sharing, download, and approval permissions shall	High

		organizational security classifications such as Public, Internal, Confidential, and Restricted.	be governed by classification levels.	
NFR-SEC-009	Cyber Incident Management	The solution shall implement security controls aligned with CERT-In incident reporting and cyber incident response guidelines, including incident detection, reporting, investigation, and recovery procedures.	Cyber incidents shall be logged, tracked, and responded to in accordance with organizational incident response procedures.	High
NFR-SEC-010	Security Monitoring	The platform shall support continuous security monitoring through centralized log collection, Security Information and Event Management (SIEM), or Security Operations Centre (SOC) integration, where applicable.	Critical security events shall generate automated alerts for investigation and response.	High
NFR-SEC-011	Privacy & Data Protection	The solution shall comply with the Digital Personal Data Protection (DPDP) Act, 2023 by implementing appropriate controls for collection, processing, storage, correction, retention, and deletion of personal data.	Personal data shall be managed throughout its lifecycle in accordance with applicable regulatory requirements.	High
NFR-SEC-012	Secure Configuration Management	The application, databases, operating systems, and supporting infrastructure shall follow secure configuration baselines and controlled change management procedures throughout the system lifecycle.	Configuration changes shall be approved, documented, logged, and periodically reviewed.	Medium
NFR-SEC-013	Compliance Verification	The solution shall support periodic verification of compliance with MeitY Cloud Guidelines, CERT-In Guidelines, GIGW 3.0, ISO/IEC 27001, and other applicable Government standards.	Compliance assessments and audit reports shall be available for review by authorized officials.	High

7.2 Usability and Accessibility Requirements

Requirement ID	Category	Requirement	Acceptance Criteria	Priority
NFR-USA-001	User Interface	The application shall provide an intuitive web-based user interface accessible through modern web browsers including Google Chrome, Microsoft Edge, and Mozilla Firefox.	All supported browsers shall render the application correctly.	High
NFR-USA-002	Role-Based Dashboards	The system shall provide personalized dashboards for Super User, Administrator, Delegation Leader, Coordinator, Convener,	Dashboard content shall be dynamically generated based on user role.	High

		Member, and Ministry Reviewer roles.		
NFR-USA-003	Collaborative Editing	The portal shall support rich-text editing with track changes, comments, and collaborative document preparation capabilities.	Multiple users shall be able to review and edit documents as per assigned permissions.	Medium
NFR-USA-004	Responsive Design	The application shall support responsive layouts for desktops, laptops, and tablets.	User interface shall automatically adapt to supported screen resolutions.	High
NFR-USA-005	User Assistance	The system shall provide context-sensitive help, tooltips, validation messages, and guidance for first-time users.	Help information shall be available throughout the application.	Medium
NFR-USA-006	Accessibility (WCAG)	The application shall comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA and GIGW 3.0 accessibility requirements to ensure usability for persons with disabilities.	Accessibility validation shall confirm compliance with WCAG 2.1 AA requirements for supported pages and user interfaces.	High
NFR-USA-007	Keyboard Accessibility	All major application functions shall be fully accessible using keyboard navigation without requiring a pointing device.	Users shall be able to navigate menus, forms, dashboards, and document workflows using keyboard controls.	High
NFR-USA-008	Screen Reader Compatibility	The application shall support commonly used screen readers through appropriate semantic markup, labels, alternative text, and accessible navigation controls.	Core application screens shall be usable through supported screen reader software.	Medium
NFR-USA-009	Consistency & Navigation	The portal shall maintain a consistent user interface, navigation structure, terminology, icons, and workflow behavior across all modules.	Users shall experience uniform navigation and interaction throughout the application.	Medium
NFR-USA-010	Search & Discoverability	The application shall provide intuitive search, filtering, sorting, and navigation mechanisms to enable users to quickly locate committees, meetings, agenda items, documents, tasks, and reports.	Users shall be able to retrieve required information using search and filtering options within acceptable response times.	High
NFR-USA-011	Localization & Date Formats	The system shall support standardized Government of India date, time, number, and document formatting conventions and shall be designed to accommodate future multilingual requirements.	System-generated content shall follow standardized formatting conventions throughout the application.	Medium

7.3 Performance and Scalability Requirements

Requirement ID	Category	Requirement	Acceptance Criteria	Priority
NFR-PER-001	Concurrent Users	The system shall support a minimum of 150 concurrent users and be scalable to support at least 300 concurrent users during peak periods.	System performance shall remain within defined SLA limits.	High
NFR-PER-002	Response Time	Common user transactions such as login, dashboard loading, document retrieval, and feedback submission shall be completed within three seconds under normal operating conditions.	Average response time shall not exceed three seconds.	High
NFR-PER-003	Document Handling	The portal shall support upload and download of documents up to 100 MB without degradation in system performance.	Upload and download operations shall complete successfully within acceptable time limits.	Medium
NFR-PER-004	Database Performance	The database shall be optimized to ensure efficient retrieval of committees, meetings, agenda items, documents, and tasks.	Query response times shall remain within acceptable performance thresholds.	High
NFR-PER-005	Scalability	The application architecture shall support horizontal scaling through containerized or microservices-based deployment without requiring significant redesign.	Additional computing resources shall be accommodated with minimal downtime.	High
NFR-PER-006	High Availability	The solution shall support high	The application shall achieve a minimum uptime of 99.5%	High

		availability through redundant application and database components to minimize service disruption.	excluding scheduled maintenance.	
NFR-PER-007	Disaster Recovery	The solution shall support Disaster Recovery (DR) with Recovery Point Objective (RPO) not exceeding 15 minutes and Recovery Time Objective (RTO) not exceeding 2 hours.	Periodic DR drills shall demonstrate compliance with defined RPO and RTO targets.	High
NFR-PER-008	Load Balancing	The application shall support load balancing across application servers to ensure optimal resource utilization and uninterrupted service during peak workloads.	Application performance shall remain within SLA limits during load balancing operations.	Medium
NFR-PER-009	Monitoring & Resource Utilization	The platform shall continuously monitor application performance, infrastructure health, database utilization, storage capacity, and network performance.	Performance dashboards and alerts shall be available for system administrators.	High
NFR-PER-010	Backup Performance	The solution shall support automated backup operations without significantly impacting application performance during business hours.	Scheduled backup jobs shall complete successfully within the defined maintenance window.	Medium
NFR-PER-011	Elastic Infrastructure	The infrastructure shall support dynamic allocation of compute, storage, and network resources	Infrastructure resources shall be scalable without requiring application redesign.	Medium

		in accordance with Government Cloud (MeghRaj) deployment principles.		
NFR-PER-012	Archival Performance	Archived meetings, documents, agenda items, and historical records shall remain searchable and retrievable without significant degradation in system performance.	Archived records shall be searchable and retrievable within acceptable response time thresholds.	Medium

7.4 Auditability and Compliance Requirements

Requirement ID	Category	Requirement	Acceptance Criteria	Priority
NFR-AUD-001	Audit Trail	The system shall maintain a complete audit trail of all significant user activities throughout the application lifecycle.	Audit records shall be searchable and retained as per the defined record retention policy.	High
NFR-AUD-002	Regulatory Compliance	The solution shall comply with applicable Government of India regulations including GFR 2017, MeitY Cloud Guidelines, ISO/IEC 27001, ISO/IEC 27017, ISO/IEC 27018, CERT-In Guidelines, DPDP Act 2023, MeghRaj Cloud Policy, NIC Application Design Guidelines, and GIGW 3.0.	Compliance documentation and evidence shall be maintained throughout the project lifecycle.	High
NFR-AUD-003	Monitoring	System errors, exceptions, security events, and operational activities shall be	Alerts shall be generated for critical events and monitored by authorized administrators.	High

		logged and monitored proactively to facilitate timely issue resolution.		
NFR-AUD-004	Data Classification	The system shall support classification of documents and records based on organizational sensitivity (e.g., Public, Internal, Confidential, Restricted) to enable appropriate security controls and access restrictions.	Every document shall be assigned to classification before publication or approval.	High
NFR-AUD-005	Record Retention	The solution shall support configurable record retention schedules for meetings, agenda items, documents, audit logs, reports, approvals, and archived records in accordance with Government record management policies.	Retention schedules shall be configurable and automatically enforced.	High
NFR-AUD-006	Audit Log Retention	Audit logs shall be retained for a configurable period in accordance with applicable Government policies and organizational requirements.	Audit logs shall remain searchable throughout the configured retention period.	High
NFR-AUD-007	DPDP Compliance	The system shall support data lifecycle management including collection, storage, access, correction, archival, retention, and deletion of personal data in	Personal data shall be processed only in accordance with applicable legal and organizational requirements.	High

		accordance with the Digital Personal Data Protection (DPDP) Act, 2023.		
NFR-AUD-008	Incident Management	The platform shall support cyber incident logging, reporting, escalation, and response in accordance with CERT-In Guidelines.	Security incidents shall be logged, tracked, and managed through defined incident response procedures.	High
NFR-AUD-009	Compliance Reporting	The system shall generate compliance reports covering audit activities, security events, workflow approvals, document access, and regulatory compliance status.	Compliance reports shall be available for authorized users and exportable in PDF and Excel formats.	Medium
NFR-AUD-010	Security Monitoring (SOC/SIEM)	The platform shall support integration with Security Operations Centre (SOC) and Security Information and Event Management (SIEM) solutions for centralized monitoring, event correlation, and security analytics.	Security events shall be available for centralized monitoring and analysis.	Medium
NFR-AUD-011	Audit Analytics	The system shall provide dashboards and analytics for monitoring user activities, workflow approvals, document access, login trends, and other audit events.	Authorized administrators shall be able to generate audit analytics and activity reports.	Medium

7.5 Availability and Reliability Requirements

Requirement ID	Category	Requirement	Acceptance Criteria	Priority
NFR-AVL-001	System Availability	The system shall maintain a minimum 99.5% availability excluding planned maintenance windows to ensure uninterrupted access to users.	Monthly system availability shall not fall below 99.5%, excluding approved maintenance periods.	High
NFR-AVL-002	Planned Maintenance	Planned maintenance activities shall be scheduled during non-business hours and communicated to all stakeholders in advance.	Planned maintenance notifications shall be issued at least 48 hours prior to scheduled downtime.	Medium
NFR-AVL-003	Backup Management	The system shall perform automated backups of application data, databases, documents, and configuration settings at predefined intervals.	Backup jobs shall execute successfully as per the configured schedule and be verified periodically.	High
NFR-AVL-004	Backup Retention	The system shall maintain backup copies in accordance with the approved backup retention policy to support operational recovery and regulatory compliance.	Backup retention shall be configurable and comply with organizational policies.	High
NFR-AVL-005	Disaster Recovery	The solution shall support Disaster Recovery (DR) capabilities through a dedicated Disaster Recovery environment hosted in accordance with Government cloud guidelines.	Disaster Recovery infrastructure shall be operational and periodically tested.	High
NFR-AVL-006	Recovery Point Objective (RPO)	The Disaster Recovery solution shall support a maximum Recovery Point Objective (RPO) of 15 minutes.	Periodic DR testing shall demonstrate compliance with the defined RPO.	High
NFR-AVL-007	Recovery Time Objective (RTO)	The Disaster Recovery solution shall support a maximum Recovery Time Objective (RTO) of 4 hours.	Periodic DR testing shall demonstrate recovery within the defined RTO.	High
NFR-AVL-008	Redundancy	The application shall eliminate single points of failure through redundancy across application servers, databases, storage, and network components.	Critical infrastructure components shall be configured with redundancy to ensure uninterrupted service availability.	High
NFR-AVL-009	Business Continuity	The platform shall support Business Continuity through documented recovery procedures, periodic	Business Continuity Plans (BCP) and Disaster Recovery Plans (DRP)	High

		Disaster Recovery drills, and continuity planning aligned with Government guidelines.	shall be tested and reviewed periodically.	
NFR-AVL-010	Health Monitoring	The system shall continuously monitor the health and availability of application services, databases, APIs, storage, and infrastructure components.	Automated alerts shall be generated for critical failures, resource exhaustion, and service interruptions.	High
NFR-AVL-011	Failover Mechanism	The solution shall support automated or controlled failover between Production and Disaster Recovery environments to minimize service disruption during failures.	Failover procedures shall be validated through periodic Disaster Recovery drills.	High
NFR-AVL-012	Data Integrity During Recovery	The system shall preserve the integrity and consistency of application data during backup, restoration, and disaster recovery operations.	Post-recovery validation shall confirm complete restoration of application data without corruption or loss.	High

7.6 Interoperability Requirements

Requirement ID	Category	Requirement	Acceptance Criteria	Priority
NFR-INT-001	Data Exchange	The system shall support import and export of reports and documents in PDF, Microsoft Excel, XML, and CSV formats.	Files shall conform to supported data formats without information loss.	High
NFR-INT-002	Open APIs	The application shall expose secure RESTful APIs to facilitate future integration with DG Shipping e-Governance applications and other national maritime digital platforms.	APIs shall support secure authentication, standardized data exchange protocols, and API documentation.	High
NFR-INT-003	Government Identity Integration	The system shall support integration with Government identity providers including NIC Single Sign-On (SSO), ePramaan, or other approved authentication services.	Users shall be able to authenticate through supported Government identity services where applicable.	High
NFR-INT-004	Metadata Exchange	The system shall support standardized metadata exchange for committees, meetings, agenda items, documents, tasks, and reports to facilitate interoperability with external systems.	Metadata shall conform to approved exchange formats without data loss.	Medium
NFR-INT-005	Document Interoperability	The platform shall support interoperability of official documents while preserving document	Imported and exported documents shall retain associated metadata and document integrity.	Medium

		metadata, version history, approval status, and digital references during import and export operations.		
NFR-INT-006	Notification Integration	The application shall support integration with Government-approved email and SMS gateways for automated notifications, alerts, reminders, and workflow communications.	Notifications shall be successfully delivered through configured communication channels.	Medium
NFR-INT-007	External System Integration	The platform shall be designed to support future integration with DG Shipping applications, NIC platforms, document management systems, and other Government e-Governance applications through secure APIs.	The solution architecture shall support configurable integration without major application redesign.	High
NFR-INT-008	Standards Compliance	All APIs and integration interfaces shall comply with Government of India interoperability standards, secure communication protocols, and open architecture principles.	Integration interfaces shall support secure, standards-based communication and authentication mechanisms.	Medium
NFR-INT-009	Search & Knowledge Integration	The system shall support integration of enterprise search capabilities across committees, meetings, agenda	Authorized users shall be able to perform unified searches across integrated repositories.	Medium

		items, documents, reports, archived records, and institutional knowledge repositories.		

8 Key Detailed Workflow Design

8.1 User Registration Workflow

8.1.1 Objective

The User Registration Workflow defines the process for onboarding users onto the IMO Strategic Engagement Portal. The workflow ensures that only authorized personnel from the Directorate General of Shipping (DGS), Ministry of Ports, Shipping and Waterways (MoPSW), Mercantile Marine Departments (MMDs), research organizations, industry associations, consultants, and other approved stakeholders are granted access to the portal. The workflow establishes a secure mechanism for user registration, role assignment, committee mapping, and account activation while maintaining appropriate governance and auditability.

8.1.2 Actors

- Super User (DDG – International Cooperation)
- Administrator
- Delegation Leader (for external member validation)
- Registered User

8.1.3 Workflow Description

The registration process begins with the identification and nomination of users who are required to participate in IMO-related activities. Depending on the category of user, the registration request may originate from the Administration, Committee Coordinator, or Delegation Leader. User information, including organization, designation, contact details, and proposed role, is entered into the system. For external organizations such as research institutions, classification societies, industry bodies, consultants, or academia, the registration request is reviewed and validated by the Delegation Leader before activation. Once approved, the Administrator creates the user account, assigns the appropriate role(s), committee memberships, and access permissions based on organizational responsibilities. The user receives login credentials or an activation link through the registered email address. Upon first login, the user is required to verify the account, change the temporary password, configure Multi-Factor Authentication (where applicable), and accept the portal's terms of use before gaining access to the assigned dashboard.
All registration activities, approvals, modifications, and account status changes are recorded in the audit log to ensure complete traceability.

The system shall maintain a centralized Stakeholder Organization Registry containing details of participating Ministries, Government Departments, Mercantile Marine Departments (MMDs), Research Institutions, Classification Societies, Industry Associations, Academic Institutions, Consultants, and other approved organizations. During user registration, every user shall be mapped to an organization to facilitate committee composition, reporting, stakeholder-wise analytics, and institutional knowledge management.

The portal shall support complete user lifecycle management including account activation, temporary suspension, role modification, transfer between committees, expiry of temporary memberships, deactivation upon completion of assignments, and account reactivation where authorized. All lifecycle events shall be subject to role-based approval and recorded in the audit trail.

The system shall also support delegated authority during periods of official absence. Where approving authority is unavailable, an Acting Officer or Delegated Authority may be assigned by the Administrator or Super User in accordance with organizational policies. Delegated permissions shall remain valid only for the approved period and shall be automatically revoked upon expiry.

8.1.4 Workflow Steps

- Identification and nomination of users.
- Submission of user details by the Administrator.
- Validation of external members by the Delegation Leader.
- Approval of registration request.
- Creation of user accounts.
- Assignment of user role(s).
- Mapping of committee membership.
- Mapping of user to the Stakeholder Organization Registry.
- Configuration of account validity period and user lifecycle status.
- Assignment of delegated authority, where applicable.
- Generation of login credentials.
- User activation and first-time login.
- Password change and MFA configuration.
- Access to role-based dashboard.
- Periodic review of user access permissions.
- Annual membership validation and renewal.

8.1.5 Business Rules

- Every user shall possess a unique User ID.
- One user may belong to multiple committees.
- Users may hold multiple roles, subject to approval.
- External users shall require Delegation Leader approval before activation.
- Only users shall be permitted to access the portal.
- All registration activities shall be recorded in the audit log.
- Every user shall be mapped to a registered Stakeholder Organization.
- User accounts shall support lifecycle states including Active, Suspended, Expired, Deactivated, and Reactivated.
- Delegated authority assignments shall be time-bound and automatically revoked upon expiry of the delegation period.
- Committee memberships and user access rights shall be reviewed and validated annually.
- Historical user activities shall remain available for audit, reporting, and compliance purposes even after account deactivation.

8.1.6 Outputs

- Active user account

- Assigned role(s)
- Committee membership
- User dashboard
- Audit trail entry
- Stakeholder organization mapping
- User lifecycle status
- Delegated authority records (where applicable)
- Membership validation status
- User activity analytics dashboard

8.2 Annual Committee Constitution Workflow

8.2.1 Objective

The Annual Committee Constitution Workflow governs the establishment and periodic reconstitution of IMO Shadow Committees and associated working structures. The workflow ensures that committee composition, leadership assignments, and stakeholder participation are updated annually to reflect current organizational requirements and areas of expertise.

8.2.2 Actors

- Super User
- Administrator
- Delegation Leader
- Committee Convener Coordinator

8.2.3 Workflow Description

At the beginning of each annual IMO cycle, the Administrator initiates the committee constitution process by creating or updating committee structures within the portal. Existing committees, including the Assembly, Council, Committees, Sub-Committees, Working Groups, and Correspondence Groups, are reviewed and updated as required.

Internal members are nominated by the Administration, while external experts from academia, research organizations, classification societies, shipping companies, ports, and other maritime stakeholders are proposed based on subject matter expertise. The Delegation Leader reviews and approves external nominations before they are included in the committee.

Following approval, the Administrator assigns committee leadership roles including Delegation Leader, Convener, Coordinator, and Members. Committee-specific permissions and access rights are automatically configured by the system. Once finalized, committee members receive automated notifications confirming their committee assignments.

The committee repository is updated and made available for the upcoming IMO meeting cycle. The system shall also maintain a Committee Charter Repository containing the Terms of Reference (ToR), committee objectives, scope, responsibilities, governance structure, and tenure of each Committee, Sub-Committee, Working Group, and Correspondence Group. Historical committee compositions, leadership changes, committee outputs, attendance records, and participation history shall be preserved to support institutional memory and facilitate future reference. The platform shall also generate committee performance indicators

such as meeting participation, agenda completion, submission timeliness, and action item completion status to assist leadership in monitoring committee effectiveness.

8.2.4 Workflow Steps

- Initiation of annual committee constitution.
- Review of existing committee structures.
- Creation or modification of committees.
- Nomination of internal members.
- Nomination of external experts.
- Validation of external members.
- Assignment of committee leadership.
- Assignment of member roles.
- Configuration of committee permissions.
- Notification to committee members
- Creation and approval of Committee Charter (Terms of Reference).
- Mapping committee members to their respective stakeholder organizations.
- Configuration of committee performance indicators and participation metrics.
- Configuration of attendance tracking for committee meetings.
- Creation of Committee Output Repository for storing committee deliverables.
- Archival of previous committee composition, leadership history, and committee outputs.

8.2.5 Business Rules

- Committee structures shall remain configurable.
- A member may participate in multiple committees.
- Committee leadership shall be assigned by the Administrator.
- External members shall require validation before inclusion.
- Committee history shall be retained for future reference.
- Every committee shall maintain an approved Committee Charter (Terms of Reference).
- Committee outputs, reports, recommendations, and deliverables shall be linked to the respective committee repository.
- Attendance and participation records shall be maintained for all committee meetings and activities.
- Committee performance shall be monitored using predefined Key Performance Indicators (KPIs).
- Committee memberships shall be reviewed and validated annually as part of the committee reconstitution process.
- Members shall be mapped to their respective stakeholder organizations for reporting and stakeholder analytics.

8.2.6 Outputs

- Updated committee repository
- Committee membership list
- Role assignment records
- Notification to committee members
- Committee Charter Repository
- Committee History Repository

- Committee Output Repository
- Committee Performance Dashboard
- Attendance and Participation Register
- Stakeholder Organization Mapping

8.3 Meeting Creation Workflow

8.3.1 Objective

The Meeting Creation Workflow defines the process for planning, scheduling, configuring, and managing IMO meetings within the portal. The workflow ensures that meetings are systematically organized with associated agenda items, participants, documentation, and timelines.

8.3.2 Actors

- Administrator
- Delegation Leader
- Coordinator
- Committee Members

8.3.3 Workflow Description

The meeting creation process begins when the Administrator or authorized Coordinator creates a new meeting within the portal by entering meeting metadata such as meeting title, committee, session number, venue, start and end dates, and meeting description.

Relevant agenda documents issued by the IMO are uploaded and linked to the meeting. Committee members, Conveners, Coordinators, and Delegation Leaders are associated with the meeting based on their committee membership.

Once the meeting has been configured, the system automatically generates meeting workspaces, document repositories, task lists, and calendar entries. Dashboard widgets are updated to display meeting readiness, pending agenda preparation, assigned tasks, and upcoming deadlines. Automated notifications are sent to all participants informing them of the newly created meeting.

8.3.4 Workflow Steps

- Creation of meeting.
- Entry of meeting metadata.
- Upload of meeting circulars and documents.
- Assignment of participants.
- Creation of meeting workspace.
- Generation of document repository.
- Calendar update.
- Dashboard update.
- Notification to participants.

8.3.5 Business Rules

- Every meeting shall possess a unique Meeting ID.
- Meetings shall belong to only one committee.
- Participants shall be selected from committee membership.
- Meeting documents shall inherit committee access permissions.
- Calendar updates shall occur automatically.

8.3.6 Outputs

- Meeting workspace
- Meeting calendar
- Participant list
- Meeting dashboard
- Meeting repository

8.4 Agenda Assignment Workflow

8.4.1 Objective

The Agenda Assignment Workflow defines the process through which agenda items received from the International Maritime Organization are distributed among committee members for analysis, document preparation, feedback, and intervention development.

8.4.2 Actors

- Delegation Leader
- Convener
- Coordinator
- Committee Members

8.4.3 Workflow Description

Following creation of the meeting, agenda items are entered into the portal either manually or through bulk upload. Each agenda item is categorized according to committee, subject area, and priority. Supporting documents, circulars, and previous interventions are linked to the corresponding agenda item.

The Delegation Leader or Convener assigns ownership of each agenda item to one or more responsible members based on their expertise. Where necessary, Working Groups, Sub-Groups, or Correspondence Groups are established to manage complex agenda items requiring collaborative preparation.

Each agenda assignment includes task ownership, timelines, related documents, approval requirements, and expected deliverables. Automated notifications are generated to inform assigned members of their responsibilities, while dashboard indicators provide real-time visibility into agenda preparation progress. Throughout the preparation period, the status of each agenda item is monitored using predefined stages such as Draft, Under Review, Consolidated, Approved, and Completed. Coordinators monitor progress and intervene where necessary to ensure timely completion.

8.4.4 Workflow Steps

- Upload of agenda items.

- Agenda categorization.
- Linking of supporting documents.
- Assignment of agenda owner.
- Creation of Working Group or Correspondence Group (if required).
- Assignment of responsible members.
- Definition of timelines.
- Notification to members.
- Monitoring of agenda preparation.
- Status updates until completion.

8.4.5 Business Rules

- Every agenda item shall possess a unique Agenda ID.
- Each agenda item shall be assigned to at least one responsible member.
- Multiple members may collaborate on a single agenda item.
- Every agenda shall have one designated owner.
- Agenda status shall be updated automatically based on workflow progression.
- Dashboard indicators reflect the latest preparation status.

8.4.6 Output

- Agenda ownership records
- Working Group assignments
- Linked documents
- Assigned tasks
- Agenda progress dashboard
- Automated notifications

8.5 Task Assignment Workflow

8.5.1 Objective

The Task Assignment Workflow defines the process for assigning, monitoring, and tracking activities related to agenda preparation, document drafting, stakeholder consultation, and meeting readiness. The workflow ensures that responsibilities are clearly allocated, deadlines are monitored, and task progress is visible to all relevant stakeholders throughout the committee lifecycle.

8.5.2 Actors

- Delegation Leader
- Convener
- Coordinator
- Committee Members
- Consultant / Support Staff

8.5.3 Workflow Description

Following the assignment of agenda items, Coordinators or Delegation Leaders create tasks linked to

specific agenda items, documents, or committee activities. Each task contains details such as task title, description, responsible individual(s), priority, due date, expected deliverables, and supporting reference documents. Tasks may be assigned to individual members or multiple users depending on the complexity of the agenda item.
Upon assignment, the system automatically generates notifications and updates the personal dashboard of each assignee. Throughout the task lifecycle, members update the status of their assigned activities, upload supporting documents, and provide progress remarks where necessary. Coordinators continuously monitor pending, completed, and overdue tasks through dedicated dashboards. The system generates automated reminders before approaching deadlines and initiates configurable escalation workflows for overdue tasks. Task completion is linked to agenda preparation status, enabling leadership to monitor overall meeting readiness through real-time dashboards.

8.5.4 Workflow Steps

- Selection of agenda item or document.
- Creation of task.
- Assignment of responsible member(s).
- Definition of priority and due date.
- Linking of supporting documents.
- Automatic notification to assignees.
- Progress updates by members.
- Automated reminders before due dates.
- Escalation of overdue tasks.
- Task completion and closure.

8.5.5 Business Rules

- Every task shall have a unique Task ID.
- Each task shall be linked to an agenda item or document.
- Multiple users may be assigned to a single task.
- Every task shall have one primary owner.
- Task priorities shall be configurable.
- Reminder frequency shall be configurable by the Administrator.
- Escalation shall occur automatically when predefined conditions are met.
- Completed tasks shall become read-only unless reopened by an authorized user..

8.5.6 Outputs

- Task assignment records
- Task progress dashboard
- Reminder notifications
- Escalation alerts
- Task completion reports

8.6 Document Preparation Workflow

8.6.1 Objective

The Document Preparation Workflow defines the process for drafting, reviewing, editing, approving, and finalizing official documents associated with IMO meetings, including working papers, information papers, interventions, delegation briefs, national positions, and supporting reference documents.

8.6.2 Actors

- Member
- Convener
- Coordinator
- Delegation Leader
- Ministry Reviewer
- Administrator

8.6.3 Workflow Description

Once agenda ownership has been established, dedicated document workspaces are automatically created for each agenda item. Members prepare draft papers using the portal's collaborative document editor or upload externally prepared documents. Each document is categorized according to its type, linked to the corresponding agenda item, and stored within the meeting repository.

The platform supports collaborative editing with version control, comments, track changes, and document comparison. Multiple contributors may work simultaneously while maintaining complete version history and user attribution. Supporting documents, reference papers, previous interventions, and IMO circulars remain accessible throughout the drafting process.

After preparation, the draft document is submitted for review through a predefined approval workflow. Reviewers provide comments and recommendations, following which revised versions are prepared until the document reaches the required level of maturity. Once approved, the document is locked to prevent unauthorized modifications while preserving all historical versions for audit purposes.

8.6.4 Workflow Steps

- Creation of document workspace.
- Draft preparation.
- Upload of supporting documents.
- Collaborative editing.
- Version management.
- Internal review.
- Revision based on comments.
- Submission for approval.
- Final approval.
- Document locking and archival

8.6.5 Business Rules

- Every document shall have a unique Document ID.
- Documents shall be linked to meetings and agenda items.
- Version history shall be maintained automatically.
- Previous versions shall remain retrievable.

- Final approved documents shall be locked.
- Only authorized users shall modify draft documents.
- All document actions shall be recorded in the audit log.

8.6.6 Outputs

- Draft documents
- Version history
- Review comments
- Approved documents
- Final document repository

8.7 National Position Development Workflow

8.7.1 Objective

The National Position Development Workflow establishes a structured process for developing India's official position on individual IMO agenda items by consolidating technical inputs, stakeholder comments, and committee deliberations into a single approved national position.

8.7.2 Actors

- Committee Members
- Convener
- Coordinator
- Delegation Leader
- Ministry Reviewer
- DG Shipping
- MoPSW

8.7.3 Workflow Description

Following document preparation, committee members review agenda papers and submit technical comments, recommendations, proposed interventions, and supporting justifications. Coordinators consolidate inputs received from members within their respective Working Groups or Correspondence Groups. Duplicate comments are merged, conflicting opinions are highlighted, and unresolved issues are flagged for further discussion.

The Delegation Leader reviews consolidated inputs from all groups and prepares the draft national position for each agenda item. Historical interventions, previous meeting decisions, international commitments, and relevant regulations remain available as reference material throughout the drafting process. Once finalized, the draft national position proceeds through the formal approval workflow before becoming the official position to be presented during the IMO meeting.

8.7.4 Workflow Steps

- Review of agenda documents.
- Submission of technical comments.
- Group-level consolidation.

- Resolution of conflicting observations.
- Preparation of draft national position.
- Review by Delegation Leader.
- Submission for approval.
- Approval and publication of final position.

8.7.5 Business Rules

- Each agenda item shall have only one approved national position.
- Historical national positions shall remain accessible.
- All stakeholder comments shall remain traceable.
- Consolidation history shall be preserved.
- Final positions shall become read-only after approval.

8.7.6 Outputs

- Draft National Position
- Consolidated comments
- Approved National Position
- National Position Repository

8.8 Feedback Consolidation Workflow

8.8.1 Objective

The Feedback Consolidation Workflow defines the process for collecting, reviewing, consolidating, and finalizing stakeholder feedback received on agenda items, draft papers, interventions, and national position documents.

8.8.2 Actors

- Member
- Convener
- Coordinator
- Delegation Leader
- Ministry Reviewer

8.8.3 Workflow Description

The workflow begins when draft documents are circulated among committee members for review. Members submit structured feedback using predefined templates or the integrated rich-text editor. Feedback may include technical observations, policy recommendations, editorial suggestions, regulatory references, or proposed modifications to draft interventions.

The coordinator reviews all submissions and categorizes comments according to agenda item, topic, contributor, and priority. Similar comments are consolidated while conflicting observations are highlighted for further discussion. The Delegation Leader subsequently reviews the consolidated feedback and prepares the committee's final response. Throughout the workflow, dashboard indicators display pending reviews, overdue submissions, consolidated feedback status, and committee readiness.

The system also maintains a historical repository of all feedback submissions, enabling future committees to reference previous discussions, understand the rationale behind decisions, and promote institutional knowledge retention.

8.8.4 Workflow Steps

- Circulation of draft documents.
- Submission of member feedback.
- Categorization of comments.
- Consolidation of feedback.
- Resolution of conflicting observations.
- Preparation of committee response.
- Review by Delegation Leader.
- Finalization of consolidated feedback.

8.8.5 Business Rules

- Feedback shall be linked to a specific agenda item or document.
- Feedback shall remain editable until submission.
- Consolidation history shall be retained.
- Duplicate comments may be merged during consolidation.
- Historical feedback shall remain searchable.
- Final consolidated feedback shall become read-only following approval.

8.8.6 Outputs

- Individual feedback records
- Consolidated feedback reports
- Committee recommendations
- Feedback dashboards
- Historical feedback repository

8.9 Approval Workflow

8.9.1 Objective

The Approval Workflow establishes a structured, transparent, and auditable process for reviewing and approving all official documents prepared through the IMO Strategic Engagement Portal. The workflow ensures that draft interventions, national positions, working papers, information papers, delegation briefs, and other official submissions undergo multiple levels of review and approval before becoming the official position of the Government of India for presentation at IMO meetings.

8.9.2 Actors

- Member
- Convener
- Coordinator
- Delegation Leader
- Ministry Reviewer (CS/NA/CSS)

- DG Shipping
- Ministry of Ports, Shipping and Waterways (MoPSW)

8.9.3 Workflow Description

Following the completion of document preparation and feedback consolidation, the document enters a predefined approval workflow. The approval hierarchy is configurable and may vary depending on the type of document, committee, or meeting. The default approval sequence shall follow the organizational hierarchy, beginning with the Member responsible for preparing the draft and progressing through the Convener, Coordinator, Delegation Leader, Ministry Reviewer, Director General of Shipping, and the Ministry of Ports, Shipping and Waterways, where applicable.

At each stage, reviewers may approve the document, return it with comments for revision, or reject it with appropriate justification. The system shall notify the next approver automatically upon successful approval of the preceding stage. Where revisions are requested, the document shall return to the previous stage while preserving the complete version history and reviewer comments.

Once the final approval is granted, the document shall be locked against further modification and designated as the official approved version. All approval actions, timestamps, comments, and workflow transitions shall be permanently recorded within the audit log.

8.9.4 Workflow Steps

- Submission of finalized draft.
- Review by Convener.
- Review by Coordinator.
- Review by Delegation Leader.
- Ministry-level review.
- DG Shipping approval.
- MoPSW approval (where applicable).
- Document locking.
- Publication of approved version.

8.9.5 Business Rules

- Approval workflows shall be configurable.
- Documents shall not bypass mandatory approval stages.
- Approvers may approve, reject, or return documents for revision.
- All reviewer comments shall remain linked to the document.
- Final approved documents shall become read-only.
- Every approval action shall be auditable.

8.9.6 Outputs

- Approved document
- Approval history
- Audit trail
- Final document repository

- Approval notifications

8.10 Meeting Execution Workflow

8.10.1 Objective

The Meeting Execution Workflow supports the live management of IMO meetings by enabling Delegation Members to access approved documents, record live interventions, capture decisions, and update documents dynamically throughout the course of the meeting.

The workflow also establishes governance mechanisms for preparing, reviewing, approving, and recording real-time interventions during live committee proceedings, ensuring that India's official position remains consistent, traceable, and appropriately authorized.

8.10.2 Actors

- Delegation Leader
- Committee Members
- Convener
- Coordinator
- Support Staff
- Ministry Reviewer (where applicable)

8.10.3 Workflow Description

Once the meeting commences, the portal transitions from preparation mode to execution mode. Approved agenda papers, interventions, briefing notes, and supporting documents become available within a dedicated Meeting Workspace. Members can quickly retrieve relevant agenda documents while discussions are underway.

Where discussions evolve differently from anticipated positions, authorized users may prepare revised intervention text within the collaborative workspace. Real-time editing, discussion boards, comments, and internal messaging facilitate coordination among delegation members without interrupting the meeting proceedings.

The system records live notes, decisions taken by the committee, amendments proposed during discussions and agreed actions. Every modification is automatically time-stamped and attributed to the contributing user. Throughout the meeting, dashboards display agenda progress, completed discussions, pending agenda items, and intervention status.

Where modifications to India's approved intervention become necessary due to developments during the meeting, the revised intervention shall follow a controlled governance process. All proposed changes shall be version controlled, attributed to the originating user, and routed to the Delegation Leader (or designated authority) for approval before being adopted as the official intervention. The platform shall maintain a complete intervention history, including draft versions, reviewer comments, approval records, timestamps, and the final intervention delivered during the meeting.

The system shall also record whether each intervention was Delivered, Partially Delivered, Not Delivered, or Withdrawn, together with reasons where applicable, thereby creating a permanent institutional record for future reference.

8.10.4 Workflow Steps

- Activation of Meeting Workspace.
- Access to approved agenda documents.
- Real-time intervention editing.
- Internal collaboration and discussions.
- Recording of meeting notes.
- Capture of committee decisions.
- Update of intervention records.
- Closure of agenda discussions
- Submission of revised intervention (where required).
- Review and approval of revised intervention by Delegation Leader.
- Recording of intervention delivery status.
- Capture of discussion outcomes and agreed follow-up actions.
- Generation of live intervention history.

8.10.5 Business Rules

- Only approved users shall access the Meeting Workspace.
- Live edits shall preserve document history.
- Every change shall record timestamp and author.
- Final interventions shall supersede draft versions.
- Meeting notes shall remain editable until meeting closure.
- Only authorized users shall modify approved interventions during live meetings.
- Every revised intervention shall maintain complete version history and approval records.
- No intervention shall overwrite previously approved versions; all revisions shall remain traceable.
- The final intervention delivered during the meeting shall be designated as the official version and preserved within the institutional repository.
- Intervention delivery status shall be captured for every agenda item.

8.10.6 Outputs

- Final interventions
- Meeting notes
- Decision register
- Updated agenda status
- Meeting activity logs
- Intervention Register
- Intervention Approval History
- Intervention Version History
- Intervention Delivery Status Report
- Live Discussion Log

8.11 Post-Meeting Reporting Workflow

8.11.1 Objective

The Post-Meeting Reporting Workflow captures the outcomes of IMO meetings and facilitates the preparation of official reports, summaries, decisions, and follow-up documentation for organizational reference and future actions.

The workflow also supports automated generation of management reports, regulatory follow-up reports, committee performance analytics, and institutional knowledge assets to improve future meeting preparedness and decision-making.

8.11.2 Actors

- Agenda Leader
- Delegation Leader
- Administrator
- Coordinator
- Committee Members
- Super User

8.11.3 Workflow Description

Upon completion of the meeting, Agenda Leaders upload the final outcome for each agenda item, including adopted amendments, resolutions, intervention summaries, and observations. The Delegation Leader reviews these submissions and validates their completeness before approving the final meeting report.
The portal automatically consolidates information from the Meeting Workspace to generate standardized reports including Minutes of Meeting (MoM), agenda-wise summaries, intervention reports, attendance records, and decision registers. Leadership dashboards are updated with meeting completion status, participation statistics, and implementation progress.

These reports become part of the permanent organizational knowledge repository and remain available for future reference.

In addition, the system shall automatically generate agenda-wise outcome reports, regulatory impact summaries, pending action reports, intervention performance reports, submission status reports, committee participation statistics, and meeting readiness analytics. Where agenda items result in amendments to IMO instruments or future correspondence activities, the system shall generate follow-up records linked to the respective agenda item.

The generated reports shall be searchable, downloadable, and available through management dashboards for future reference, performance monitoring, and institutional learning.

8.11.4 Workflow Steps

- Submission of agenda outcomes.
- Upload of supporting documents.
- Validation by Delegation Leader.
- Generation of meeting reports.
- Approval of final report.

- Publication to repository.
- Generation of agenda-wise outcome reports.
- Generation of intervention performance reports.
- Generation of regulatory follow-up reports.
- Update of committee performance dashboards.
- Publication to knowledge repository.

8.11.5 Business Rules

- Reports shall be generated automatically where possible.
- Manual edits shall be tracked.
- Approved reports shall be immutable.
- Reports shall remain searchable.
- Export formats shall include PDF, Excel, and XML.
- Every agenda item shall have an associated outcome record.
- Regulatory amendments and future actions arising from meetings shall be linked to corresponding agenda items.
- Generated reports shall be retained in accordance with the document retention policy.
- Only approved reports shall be published to the organizational repository.
- All report modifications and approvals shall be captured in the audit trail.

8.11.6 Outputs

- Minutes of Meeting
- Agenda-wise Summary Report
- Intervention Report
- Attendance Register
- Decision Register
- Institutional Knowledge Repository Update

8.12 Action Item Closure Workflow

8.12.1 Objective

The Action Item Closure Workflow ensures systematic monitoring and closure of all post-meeting commitments, regulatory actions, follow-up activities, and implementation responsibilities arising from IMO meetings.

The workflow also facilitates tracking of regulatory amendments, implementation of IMO decisions, correspondence group commitments, submission obligations, and long-term follow-up activities until formal closure by the competent authority.

8.12.2 Actors

- Administrator
- Delegation Leader
- Convener
- Coordinator
- Committee Members

- Ministry Reviewer
- Super User

8.12.3 Workflow Description

Following the conclusion of the meeting, the portal automatically generates Action Items based on recorded decisions, adopted resolutions, pending regulatory amendments, and follow-up commitments. Each action item is assigned to the appropriate Convener, Coordinator, or Committee Member together with defined timelines and expected deliverables.

Assigned users periodically update progress through the portal. Coordinators monitor implementation status while Delegation Leaders review overall progress using dedicated dashboards. Automated reminders are generated before deadlines, and overdue actions are escalated in accordance with predefined business rules.

Upon completion of all required activities, supporting evidence may be uploaded before the authority responsible formally closes the action item. Closed actions remain archived for future reporting and compliance verification.

The system shall categorize action items based on activity type such as regulatory amendments, policy decisions, correspondence group submissions, future IMO submissions, committee decisions, implementation activities, and administrative follow-up.

Each regulatory action shall remain linked to the originating Committee, Meeting ID, Agenda ID, IMO Resolution, Circular Letter, Working Paper, or Information Paper to ensure complete traceability throughout its lifecycle.

The platform shall continuously monitor action ageing, SLA compliance, pending approvals, implementation progress, and regulatory readiness through dedicated management dashboards. Escalation notifications shall be generated automatically for overdue actions until formal closure.

Completed regulatory actions shall undergo verification by the designated authority before being marked as closed. The system shall preserve implementation history and supporting evidence for future reference and audit purposes.

8.12.4 Workflow Steps

- Automatic creation of action items.
- Assignment of responsibilities.
- Definition of timelines.
- Progress updates.
- Reminder generation.
- Escalation of overdue actions.
- Upload of completion evidence.
- Approval of closure.
- Classification of regulatory and non-regulatory action items.
- Linkage of action items with Agenda ID, Meeting ID and IMO documents.
- Monitoring of SLA compliance and action ageing.
- Verification of regulatory implementation.
- Closure approval by competent authority.

- Archival of completed action records.

8.12.5 Business Rules

- Every action item shall possess a unique Action ID.
- Action items shall remain open until formally approved.
- Progress shall be updated periodically.
- Escalation shall occur automatically for overdue actions.
- Closed actions shall not be modified without authorization
- Every action item shall be assigned an owner, priority level, due date, and current status.
- Regulatory follow-up actions shall remain open until implementation is verified and approved.
- Action ageing shall be automatically calculated based on predefined SLA timelines.
- The system shall generate automated reminders prior to SLA breach and escalation notifications after breach.
- Supporting evidence shall be mandatory before closure of regulatory action items.
- Complete history of status changes, approvals, comments, and closure activities shall be maintained within the audit trail.

8.12.6 Outputs

- Action register
- Progress dashboard
- Closure reports
- Compliance status
- Reminder notifications
- Regulatory Follow-up Register
- Action Ageing Dashboard
- SLA Compliance Report
- Implementation Status Dashboard
- Verified Action Closure Report
- Regulatory Compliance Repository

8.13 Archival Workflow

8.13.1 Objective

The Archival Workflow defines the long-term preservation and management of all committee records, documents, interventions, reports, meeting outcomes, and historical knowledge generated through the IMO Strategic Engagement Portal.

The workflow also establishes a centralized institutional knowledge repository to facilitate knowledge reuse, historical reference, regulatory traceability, and informed decision-making for future IMO meetings and other international engagements.

8.13.2 Actors

- Administrator
- Super User
- Delegation Leader

- Authorized Users
- Coordinator
- Committee Members (Read-only Access)

8.13.3 Workflow Description

After completion of the meeting cycle and closure of all associated action items, the portal automatically initiates the archival process. All approved documents, meeting reports, agenda records, interventions, feedback, correspondence, approvals, and audit records are transferred to the centralized knowledge repository.

Archived information is indexed using metadata such as committee, meeting, agenda number, year, document type, contributor, keywords, and subject classification to facilitate efficient search and retrieval. Previous interventions, national positions, committee reports, and regulatory decisions remain available as reference material for future meetings.

Retention policies shall govern the preservation, archival, and eventual disposal of records in accordance with organizational policies and applicable Government of India record management guidelines. Archived records shall remain protected through role-based access control while ensuring long-term institutional knowledge preservation.

The repository shall maintain complete historical records of committee constitutions, meeting outcomes, agenda deliberations, correspondence group activities, working papers, information papers, interventions, national positions, regulatory amendments, committee outputs, and follow-up actions.

The system shall provide enterprise-wide search capabilities using metadata, keywords, document numbers, agenda numbers, committee names, meeting sessions, IMO circular references, Working Paper numbers, Information Paper numbers, contributors, and document classifications. OCR-based indexing may be used for scanned documents to improve searchability.

Historical records shall be linked with subsequent meetings to enable users to trace the evolution of discussions, India's positions, regulatory amendments, and implementation history across multiple IMO sessions. The repository shall thereby serve as the institutional memory of the Directorate General of Shipping for international maritime engagements.

8.13.4 Workflow Steps

- Completion of meeting cycle.
- Verification of document completeness.
- Transfer to archive repository.
- Metadata generation.
- Indexing and cataloguing.
- Application of retention policy.
- Knowledge repository update.
- Availability for future reference.
- Classification of archived documents.
- Linking of historical records with future meetings.
- Indexing using mandatory metadata and OCR (where applicable).
- Creation of institutional knowledge assets.
- Periodic archival integrity verification.

8.13.5 Business Rules

- Only approved documents shall be archived.
- Archive records shall remain searchable.
- Metadata shall be mandatory for all archived records.
- Historical versions shall be retained.
- Archive access shall follow Role-Based Access Control.
- Record retention shall comply with applicable organizational policies.
- Every archived document shall be assigned a document classification and retention category.
- Historical interventions and national positions shall remain linked to their respective agenda items and meetings.
- Archived records shall support full-text search, metadata search, and document number search.
- Records shall be retained and disposed of in accordance with Government Record Management Policies and applicable statutory requirements.
- Only authorized users shall restore archived records for operational use.
- All archival, restoration, and disposal activities shall be recorded in the audit trail

8.13.6 Outputs

- Centralized archive repository
- Searchable knowledge base
- Historical intervention repository
- Committee history
- Meeting archive
- Long-term document repository
- Institutional Knowledge Repository
- Regulatory History Repository
- National Position Repository
- Committee Output Repository
- Searchable Historical Dashboard
- Archive Audit Logs

9 Cyber Security and Compliance Framework

The IMO Strategic Engagement Portal shall be designed and implemented with a comprehensive cyber security framework to safeguard the confidentiality, integrity, and availability of organizational information throughout the system lifecycle. The platform shall adopt a security-by-design approach, incorporating preventive, detective, and corrective security controls across the application, infrastructure, database, and network layers. Security controls shall be integrated into all stages of the software development lifecycle, including design, development, testing, deployment, operations, and maintenance.

The solution shall comply with the applicable cybersecurity standards, Government of India guidelines, and industry best practices to ensure secure processing of sensitive committee documents, national positions, stakeholder information, and official correspondence

9.1 MeitY Compliance (Security)

The application shall comply with the infrastructure, security, and cloud governance requirements prescribed by the Ministry of Electronics and Information Technology (MeitY) for Government digital platforms. The solution shall be hosted exclusively on a MeitY-empanelled Cloud Service Provider (CSP) under the MeghRaj Cloud Initiative and shall adhere to the applicable cloud security guidelines, data protection principles, and infrastructure security controls.

The deployment architecture shall comprise four logically segregated environments, namely Development, User Acceptance Testing (UAT), Production, and Disaster Recovery (DR). Each environment shall maintain independent access controls, security configurations, and administrative boundaries to ensure operational isolation, secure application lifecycle management, and protection of sensitive information across all stages of deployment.

The solution architecture shall be aligned with the prescribed Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirements to ensure continuity of critical services in the event of system disruptions. These objectives shall be achieved through robust infrastructure planning, encrypted backup mechanisms, secure storage of backup data, and periodic Disaster Recovery (DR) drills conducted after Go-Live to validate recovery procedures, demonstrate recovery readiness, and ensure business continuity under disaster scenarios.

The solution shall also comply with Government-prescribed backup, disaster recovery, business continuity, and operational monitoring requirements throughout the project lifecycle.

9.1.1. Cloud Hosting Compliance

Requirement	Measurable Compliance Criteria
Cloud Platform	Application shall be hosted only on a **MeitY empanelled Cloud Service Provider (**CSP).
Data Residency	100% of application data, backups, logs, and disaster recovery data shall reside within India.
Availability Zones	Primary and DR environments shall be hosted in geographically separate Indian data centres.
Multi-Tenancy	Logical segregation of tenant data shall be demonstrated and validated during security testing.
Infrastructure Provisioning	Infrastructure shall be provisioned through Infrastructure as Code (IaC) where feasible and maintained under version control.

9.1.2. Availability Requirements

KPI	Target
Application Availability	≥99.9% monthly (excluding approved maintenance)
Database Availability	≥99.95%
API Availability	≥99.9%
Network Availability	≥99.95%

Maintenance Window	Pre-approved and preferably outside business hours

9.1.3. Disaster Recovery

Requirement	Target
Recovery Point Objective (RPO)	≤15 minutes (critical data)
Recovery Time Objective (RTO)	≤4 hours
DR Drill	At least twice annually
Backup Verification	Monthly restoration testing
Automatic Failover	For critical production services where technically feasible

9.1.4. Information Security

The cloud environment shall demonstrate compliance with:

- ISO/IEC 27001
- ISO/IEC 27017
- ISO/IEC 27018
- CERT-In Directions
- Government of India Cyber Security Guidelines

Measurable controls include:

Requirement	Compliance Target
Encryption at Rest	AES-256 or equivalent
Encryption in Transit	TLS 1.2 or higher (TLS 1.3 preferred)
Multi-Factor Authentication	Mandatory for privileged users
Privileged Access Review	Quarterly
Security Patch Compliance	Critical patches within 7 days; High severity within 15 days
Vulnerability Assessment	Quarterly
Penetration Testing	At least annually and after major releases

9.1.5. Identity and Access Management

Requirement	Measurable Criteria

Role-Based Access Control	100% of users assigned predefined roles
Least Privilege	Privileged access documented and approved
Password Policy	Compliant with Government security policy
Single Sign-On (SSO)	Supported using SAML/OAuth/OpenID Connect where required
Inactive Account Disablement	Within 30 days of inactivity (or as per organizational policy)

9.1.6. Audit and Logging

Requirement	Target
Audit Coverage	100% of user authentication, authorization, configuration, and data modification events
Log Retention	Minimum 180 days online; archival as per record retention policy
Time Synchronization	NTP synchronization across all servers
Tamper Protection	Audit logs protected against modification
Security Monitoring	Continuous monitoring with alerting for critical events

9.1.7. Data Protection

Requirement	Target
Backup Frequency	Daily full backup with incremental backups as required
Backup Retention	Configurable to business and legal requirements
Personally Identifiable Information (PII)	Protected through encryption and access controls
Data Deletion	Secure deletion aligned with retention policy
Data Integrity	Checksum or equivalent validation for critical datasets

9.1.8. Network Security

Requirement	Target
Network Segmentation	Separate production, testing, and development environments
Web Application Firewall	Mandatory for internet-facing applications
DDoS Protection	Enabled for public endpoints
Intrusion Detection/Prevention	Active monitoring and alerting
Firewall Rule Review	Quarterly

9.1.9. Application Security

Requirement	Target
Secure Development Lifecycle (SDLC)	Mandatory
Static Code Analysis	Before each production release
Dynamic Application Security Testing (DAST)	Prior to production deployment
Dependency Scanning	Automated in CI/CD pipeline
Secrets Management	No hard-coded credentials; use secure vaults

9.1.10. API Security

Requirement	Target
Authentication	OAuth 2.0 or equivalent
API Encryption	HTTPS only
Rate Limiting	Configured for public APIs
API Logging	100% request and response logging for critical APIs
API Versioning	Mandatory

9.1.11. Monitoring and Operations

Requirement	Target
Infrastructure Monitoring	24×7
Application Monitoring	24×7
Automated Alerts	Critical alerts within 5 minutes
Incident Response	Initial response within 30 minutes for critical incidents
Root Cause Analysis	Completed within 5 working days for major incidents

9.1.12. Performance

KPI	Target
Average Page Load Time	≤3 seconds under normal load
API Response Time	≤2 seconds for 95% of requests

Concurrent Users	Support the defined peak load with acceptable performance
Auto Scaling	Enabled for production workloads where appropriate

9.1.13. Compliance and Governance

Requirement	Target
Security Compliance Review	Annual
Third-Party Risk Assessment	Annual
Configuration Review	Quarterly
Change Management	100% of production changes documented and approved
Compliance Reporting	Quarterly compliance reports to the project owner

9.1.14. Service Level Agreements (SLA)

Parameter	Target
Critical Incident Response	≤30 minutes
High Severity Incident Resolution	≤4 hours
Medium Severity Resolution	≤1 business day
Low Severity Resolution	≤5 business days
Planned Maintenance Notice	At least 7 days in advance

9.2 Data Lifecycle Management and DPDP Compliance Requirements

9.2.1 Data Classification

Data Category	Examples	DPDP Applicability	Retention Basis
Personal Data	Name, email, mobile number, designation	Yes	Business and legal purpose
Official Data	Meeting records, project documents, reports	No (unless linked to identifiable individuals)	Record retention policy
Public Data	IMO publications, circulars, public notices	No	Permanent or as required
System Data	Audit logs, access logs, security logs	May include personal identifiers	Security policy
Metadata	Login timestamps, IP addresses,	Yes, if linked to an individual	Security and audit

	browser details		requirements

9.2.2 Data Collection Controls

Requirements

- Collect only the minimum personal data necessary for the stated purpose (data minimization).
- Record the lawful purpose for every category of personal data collected.
- Configure mandatory and optional fields separately.
- Prevent collection of unnecessary sensitive information.
- Publish a clear privacy notice explaining the purpose of processing.

Measurable Compliance

- 100% of personal data fields mapped to documented processing purposes.
- Privacy notice displayed before data collection.
- Consent captured electronically where required.

9.2.3 Data Accuracy and Correction

Ensure that personal data remains accurate, complete, and up to date.

Requirements

- Allow authenticated users to review their personal information.
- Provide mechanisms to request correction or update of inaccurate information.
- Maintain version history of changes.
- Record the user, timestamp, and reason for each modification.

9.2.4 Measurable Controls

Requirement	Target
Correction Request Processing	Within 15 working days
Audit Trail	100% of modifications logged
Version History	Retained for all corrections
Validation	Mandatory for critical fields before saving

9.2.5 Data Retention Policy

Data shall not be retained longer than necessary for the purpose for which it was collected, unless required by law or Government record-retention policies.

Data Type	Suggested Retention
User Profiles	Active period plus 3 years after account closure

Authentication Logs	Minimum 180 days online; archive according to organizational policy
Security Logs	1–3 years, based on risk and audit needs
Project Documents	As per Government record retention schedule
Meeting Records	Project duration plus 5 years (or organizational policy)
Analytics Data	Until business purpose is fulfilled, then anonymize or archive

9.2.6 Measurable Controls

- Automated retention schedules configured.
- Monthly review of records exceeding retention periods.
- Retention policy approved by the Data Protection Officer (or designated authority).

9.2.7 Data Archival

Requirements

- Archive records no longer required for day-to-day operations but retained for legal, audit, or historical purposes.
- Encrypt archived data.
- Restrict access to authorized personnel.
- Maintain an archive inventory and retrieval process.

9.2.8 Compliance Targets

Requirement	Target
Archive Encryption	AES-256 or equivalent
Archive Access Logging	100%
Archive Integrity Verification	Quarterly
Archive Recovery Testing	Annual

9.2.9 Data Deletion and Disposal

Requirements

When the purpose of processing has been fulfilled and no legal obligation exists to retain the data:

- Securely delete personal data from production systems.
- Delete or overwrite backup copies according to the backup lifecycle.
- Ensure deletion is irreversible.
- Record deletion activities for audit.

Measurable Controls

Requirement	Target
Deletion Request Processing	Within 30 days, subject to legal obligations
Secure Deletion	NIST SP 800-88 or equivalent method
Deletion Audit Logs	100%
Backup Deletion	As per backup retention schedule

9.2.10 Data Anonymization

Where long-term analytics or reporting is required:

- Remove or irreversibly anonymize personal identifiers.
- Prevent re-identification through technical controls.
- Use anonymized datasets for trend analysis, dashboards, and AI models whenever possible.

Compliance Target

- 100% of archived analytics datasets anonymized unless identification is legally required.

9.2.11 Data Subject Rights (DPDP Compliance)

The platform shall support requests to:

- Access personal data.
- Correct inaccurate or incomplete personal data.
- Erase personal data when legally permissible.
- Withdraw consent (where consent is the basis for processing).
- Raise grievances through a designated mechanism.

Service Levels

Request Type	Target Resolution
Access Request	Within 15 working days
Correction Request	Within 15 working days
Erasure Request	Within 30 working days (subject to legal obligations)
Grievance	Initial response within 7 working days

9.2.12 Audit and Traceability

The platform shall maintain immutable audit logs for:

- Data creation
- Data updates

- Corrections
- Access events
- Downloads
- Exports
- Deletions
- Administrative actions

Minimum Audit Fields

- User ID
- Timestamp
- Action performed
- Source IP
- Device information (where appropriate)
- Record identifier
- Previous value (for updates)
- New value (for updates)

9.2.13 Access Control

- Role-Based Access Control (RBAC).
- Least-privilege principle.
- Multi-factor authentication for privileged users.
- Periodic access reviews (at least quarterly).
- Automatic session timeout after inactivity.

9.2.14 Security Controls

- Encryption of personal data at rest and in transit.
- Regular vulnerability assessments.
- Annual penetration testing.
- Continuous monitoring for unauthorized access.
- Security incident response procedures aligned with CERT-In requirements.

9.2.15 Compliance Monitoring

The platform provide a compliance dashboard with indicators such as:

KPI	Target
Records within retention period	100%
Records exceeding retention	0%
Pending correction requests	0 beyond SLA
Pending deletion requests	0 beyond SLA

Successful deletion verification	100%
Audit log coverage	100%
Encrypted personal data	100%
Quarterly access reviews completed	100%
Privacy impact assessments completed	100% for major new features

9.2.16 Governance

Department should assign clear responsibilities:

- **Data Fiduciary (Project Owner):** Overall accountability for compliance.
- **System Administrator:** Technical implementation of lifecycle controls.
- **Information Security Officer:** Security monitoring and incident response.
- **Data Protection Officer / Designated Grievance Officer:** Handling data subject requests and compliance oversight.
- **Business Owner:** Approving retention schedules and validating business purpose.

9.3 CERT-In Guidelines

The portal shall comply with the cybersecurity practices and incident management guidelines issued by the Indian Computer Emergency Response Team (CERT-In). Compliance shall include, but not be limited to:

- Secure application development practices.
- Continuous monitoring of security events.
- Logging of critical security incidents.
- Timely identification and reporting of cyber incidents.
- Periodic vulnerability assessments.
- Secure configuration management.
- Regular patch management and software updates.
- Incident response procedures.

Appropriate mechanisms shall be established for monitoring application security, detecting suspicious activities, and supporting forensic investigations whenever required

CERT-In Compliance
To establish a standardized process for identifying, responding to, recovering from, and reporting cyber security incidents affecting the IMO Shadow Portal while ensuring compliance with CERT-In reporting requirements

Incident Classification

Severity	Description	Response Time
Low	Minor security event	Within 8 hours
Medium	Limited impact	Within 4 hours
High	Major application impact	Within 2 hours
Critical	Data breach/service outage	Immediate

CERT-In Reporting

If the incident falls under the reportable categories specified by CERT-In (such as unauthorized access, website defacement, malware attacks, data breaches, or denial-of-service attacks), it must be reported **within 6 hours** of noticing the incident or being informed of it. Organizations must also designate a Point of Contact (PoC), maintain ICT logs for **180 days** within India, and provide information or assistance requested by CERT-In.

Typical reporting information includes:

- Organization name
- Point of Contact
- Incident date/time
- Incident type
- Affected systems
- Impact assessment
- Current status
- Mitigation actions
- Supporting logs/evidence

Roles & Responsibilities

Role	Responsibility
Application Support Team	Detect and report incidents
Security Operations Team	Analyze and contain incidents
Infrastructure Team	Server isolation and recovery
Development Team	Fix application vulnerabilities
Database Administrator	Database investigation and restoration
CERT-In Point of Contact	Regulatory reporting and coordination
Project Manager	Communication and stakeholder coordination

Management	Decision making and approvals

9.4 GIGW 3.0 Acceptance Criteria and Compliance Checklist

The IMO Strategic Engagement Platform shall comply with the Guidelines for Indian Government Websites and Applications (GIGW) Version 3.0, issued by the National Informatics Centre (NIC) under the Ministry of Electronics and Information Technology (MeitY), to ensure accessibility, usability, security, interoperability, multilingual support, and citizen-centric service delivery.

The successful bidder shall demonstrate compliance through documentation, testing, and independent verification prior to User Acceptance Testing (UAT) and Go-Live.

9.4.1. Acceptance Criteria

The application shall be accepted only if all mandatory GIGW 3.0 requirements are fully implemented and verified.

Acceptance Parameter	Success Criteria
Overall GIGW Compliance	100% compliance with all mandatory GIGW 3.0 controls
Accessibility Compliance	Conformance with WCAG 2.1 Level AA (or higher if mandated)
Responsive Design	Fully functional on desktop, tablet, and mobile devices
Browser Compatibility	Supported across current versions of major browsers
Multilingual Support	English and Hindi at minimum, with provision for additional Indian languages
Search Functionality	Site-wide search returns accurate and relevant results
Security Compliance	HTTPS enforced with valid certificates and secure configuration
Metadata Compliance	Required metadata implemented for all public pages
Performance	Page load time within defined SLA under normal operating conditions
Broken Links	Zero broken internal links at production release
Content Quality	Content approved by designated business owner
Privacy & Legal Notices	Published and accessible from all public pages
Accessibility Testing	Successful completion using automated tools and manual validation
Security Testing	No unresolved Critical or High vulnerabilities before Go-Live

B. Performance

KPI	Target
Homepage load time	≤3 seconds
Dashboard load time	≤5 seconds
Search response	≤3 seconds
API response	≤2 seconds for 95% of requests
Concurrent users	Supports approved peak load

C. Security

Requirement	Acceptance Criteria
HTTPS enforced	Yes
TLS 1.2 or higher	Yes (TLS 1.3 preferred)
Secure cookies	Implemented
Session timeout	Configured
Password policy	Implemented
Multi-factor authentication (privileged users)	Enabled
Audit logging	Enabled
Vulnerability Assessment	Completed
Penetration Testing	Completed
Critical vulnerabilities	None unresolved

D. Content Management

Requirement	Acceptance Criteria
Content ownership defined	Yes
Review workflow implemented	Yes
Version control maintained	Yes
Last updated date displayed	Yes
Metadata configured	Yes
Broken link validation	Passed

E. Multilingual Support

Requirement	Acceptance Criteria
English interface	Available
Hindi interface	Available
Language switch	Functional
Unicode support	Verified
Consistent terminology	Verified

F. Search Engine Optimization (SEO)

Requirement	Acceptance Criteria
Metadata tags	Implemented
XML sitemap	Available
Robots.txt	Configured
Descriptive page titles	Implemented
Semantic HTML	Verified

G. Interoperability

Requirement	Acceptance Criteria
API documentation	Available
Open standards used	Verified
Standard data formats (JSON/XML)	Supported
Integration logging	Enabled
Error handling	Standardized

H. Privacy and Legal Compliance

Requirement	Acceptance Criteria
Privacy Notice	Published
Cookie Notice (if applicable)	Published
Data Protection Statement	Published
Grievance contact	Published
Disclaimer	Published

I. Monitoring and Operations

Requirement	Acceptance Criteria
Uptime monitoring	Enabled
Error monitoring	Enabled
Security monitoring	Enabled
Backup verification	Completed
Disaster Recovery testing	Successful

9.5 WCAG 2.1 Level AA Accessibility Compliance

The IMO Strategic Engagement Platform shall conform to the Web Content Accessibility Guidelines (WCAG) 2.1 Level AA to ensure that persons with disabilities can perceive, operate, understand, and interact with all public-facing and authenticated user interfaces.

Measurable Accessibility Requirements

Accessibility Requirement	Measurable Acceptance Criteria
WCAG Compliance	The application shall demonstrate **100% compliance with all applicable WCAG 2.1 Level AA Success Criteria**.
Keyboard Accessibility	**100%** of interactive functions shall be operable using only a keyboard without requiring a mouse.
Keyboard Focus	All interactive elements shall display a visible keyboard focus indicator throughout navigation.
Alternative Text	**100%** of meaningful images, icons, charts, and graphical controls shall include descriptive alternative text. Decorative images shall use null (alt="") text.
Colour Contrast	Text and images of text shall maintain a minimum contrast ratio of **4.5:1** (normal text) and **3:1** (large text) as required by WCAG 2.1 AA.

Resize Text	Users shall be able to resize text up to **200%** without loss of information or functionality.
Reflow	Content shall reflow correctly at a viewport width equivalent to **320 CSS pixels** without requiring horizontal scrolling (except where permitted by WCAG).
Screen Reader Compatibility	All pages shall be compatible with standard screen readers (e.g., NVDA, JAWS, VoiceOver) without blocking access to content or functionality.
Form Accessibility	**100%** of form controls shall have programmatically associated labels, instructions, and accessible error messages.
Error Identification	Validation errors shall clearly identify the affected field and provide guidance for correction without relying solely on colour.
Page Titles	Every page shall have a unique and descriptive page title that accurately reflects its purpose.
Heading Structure	Pages shall use a logical and hierarchical heading structure (H1 to H6) without skipping levels in a way that impairs navigation.
Link Purpose	The purpose of every hyperlink shall be understandable from its text alone or from its surrounding context.
Skip Navigation	Every page shall provide a "Skip to Main Content" mechanism accessible via keyboard.
Language Declaration	Each page shall declare the primary language programmatically, and language changes within content shall be identified where applicable.
Time Limits	Where session timeouts exist, users shall receive advance notification and be provided with an option to extend the session, where security policies permit.
Multimedia Accessibility	Pre-recorded multimedia containing speech shall include synchronized captions. Audio-only content shall include an equivalent text transcript.
Responsive Accessibility	Accessibility shall be maintained across desktop, tablet, and mobile devices without loss of content or functionality.
Non-Text Controls	Buttons, icons, and other non-text controls shall have accessible names that accurately describe their function.
Accessibility Testing	Prior to Go-Live, the application shall achieve **zero unresolved Critical or High accessibility defects** identified through automated scanning and manual accessibility testing against WCAG 2.1 Level AA.

9.6 ISO 27001 Controls

The solution shall implement security controls aligned with the principles of ISO/IEC 27001 Information Security Management System (ISMS) to ensure systematic protection of information assets.
The security framework shall incorporate controls relating to:

- Information security governance.
- User identity and access management.

- Asset management.
- Secure authentication and authorization.
- Cryptographic controls.
- Physical and environmental security.
- Operations security.
- Communications security.
- Supplier relationship management.
- System acquisition, development, and maintenance.
- Information security incident management.
- Business continuity management.
- Compliance monitoring.

These controls shall be incorporated throughout the application lifecycle to minimize security risks and ensure continuous protection of organizational information.

9.7 Data Encryption Standards

The portal shall implement industry-standard cryptographic mechanisms to protect sensitive information during storage and transmission.

The encryption framework shall include:

- Encryption of all communication channels using HTTPS with TLS 1.3 or higher.
- Encryption of sensitive application data stored within databases and backup repositories.
- Secure storage of user credentials using industry-standard hashing algorithms.
- Encryption of backup files prior to storage.
- Secure key management practices.
- Protection of confidential documents during upload, download, and archival.

The encryption mechanisms shall ensure that sensitive information remains protected against unauthorized disclosure, modification, or interception.

9.8 Security Audit Requirements

The application shall undergo comprehensive security assessments throughout the project lifecycle to validate compliance with prescribed security standards and identify potential vulnerabilities. Compliance with the applicable security standards and regulatory guidelines shall be achieved through the implementation of appropriate technical, administrative, and operational security controls across the design, development, deployment, operations, and maintenance phases of the project. Security compliance shall therefore be demonstrated through effective implementation, continuous governance, monitoring, and operational practices rather than documentation alone.

The following mandatory security audits shall be conducted during the project lifecycle:

- Pre-Go-Live Security Audit
- Mid-Term Security Audit
- Final Security Audit

The scope of these audits shall include:

- Application security assessment.
- Infrastructure security review.
- Database security assessment.
- Network security verification.
- User access control validation.
- Configuration review.
- Audit log verification.
- Compliance verification.

Any critical or high-severity observations identified during these audits shall be rectified prior to acceptance of the security compliance certificate. Audit findings shall be documented, tracked, and formally closed before production deployment or project acceptance, thereby ensuring that the application continues to satisfy the required security, compliance, and operational standards throughout its lifecycle.

9.9 Vulnerability Assessment and Penetration Testing

The portal shall undergo periodic Vulnerability Assessment and Penetration Testing (VAPT) to evaluate the effectiveness of implemented security controls and identify potential weaknesses before they can be exploited.

The VAPT process shall include:

- Automated vulnerability scanning.
- Manual penetration testing.
- Authentication and authorization testing.
- Input validation testing.
- API security testing.
- Session management testing.
- Configuration assessment.
- Business logic validation.
- Security regression testing following major releases.

The assessment shall cover the application layer, infrastructure, databases, APIs, and supporting services. Identified vulnerabilities shall be classified according to their severity, assigned for remediation, and verified through re-testing before closure.

9.10 Log Management and Monitoring

The portal shall maintain comprehensive system logs and audit records to support operational monitoring, security analysis, regulatory compliance, and forensic investigations.

The logging framework shall capture events including:

- User authentication and logout.
- Failed login attempts.
- User registration and account modifications.
- Document upload, download, modification, and deletion.
- Workflow approvals and rejections.
- Task assignments and status changes.
- Administrative activities.
- Configuration modifications

- System errors and exceptions.
- Security-related events.

Log records shall include:

- User Identification
- Date and Time
- IP Address
- Module Accessed
- Activity Performed
- Result of Activity

Access to audit logs shall be restricted to authorized administrators only. Logs shall be protected against unauthorized modification and retained in accordance with organizational retention policies.

Continuous monitoring mechanisms shall support proactive identification of abnormal activities, unauthorized access attempts, application errors, and infrastructure issues. Alerts may be generated for predefined security events to facilitate timely investigation, response, and continuous improvement of the overall security posture.

9.11 Compliance Summary

The cyber security framework for the IMO Strategic Engagement Portal shall align with the following standards, regulations, and guidelines

Standard / Guideline	Compliance Objective
MeitY Cloud Guidelines	Secure Government cloud deployment and infrastructure compliance
CERT-In Guidelines	Cybersecurity incident management, monitoring, and vulnerability management
ISO/IEC 27001	Information Security Management System (ISMS) controls
ISO/IEC 27017	Cloud Security Controls
ISO/IEC 27018	Protection of Personally Identifiable Information (PII) in cloud environments
Digital Personal Data Protection (DPDP) Act, 2023	Protection and lawful processing of personal data
OWASP Top 10	Secure application development and protection against common web vulnerabilities
GIGW -3.0	The Guidelines for Indian Government Websites (GIGW) are a comprehensive set of standards established by the National Informatics Centre (NIC)

10 Data Migration Strategy

10.1 Historical/ Legacy Data Migration

The system shall support migration of historical data from identified legacy systems into the I.M.O Strategic Engagement Platform. The migrated data shall include:

- Stakeholder master data
- Engagement history (meetings, calls, interactions)
- Strategic initiatives, programs, and campaigns
- Tasks, action items, and follow-ups
- Communication records (emails, notes)
- Attachments and supporting documents

10.2 Data Extraction

- The system shall support extraction of data from multiple source systems in structured formats (CSV, database export, APIs).
- The system shall ensure completeness and consistency of extracted data.
- The system shall allow extraction based on configurable filters (e.g., date range, entity type).

10.3 Data Transformation

The system shall transform legacy data to align with I.M.O platform structure:

- Format standardization (date, time, codes)
- Value mapping and normalization
- Derived values based on business rules

The system shall preserve engagement timelines and relationships during transformation.

10.4 Data Cleansing

- The system shall support data cleansing to improve data quality:
    - Removal of duplicate stakeholder records
    - Correction of incomplete or inconsistent data
    - Standardization of naming conventions
- The system shall generate data quality reports prior to migration.

10.5 Data Integrity

- The system shall ensure referential integrity between related entities (e.g., stakeholder–engagement–activity).
- The system shall preserve chronological order of engagements.
- The system shall prevent duplication during migration.

10.6 Data Validation

The system shall validate data before and after migration:

- Mandatory field validation
- Data type validation
- Business rule validation

The system shall flag errors and exceptions for correction.

a) Data Quality Checks

The system shall perform data quality checks:

- Duplicate detection
- Referential integrity validation
- Business rule validation

The system shall generate validation reports highlighting issues.

b) Pre-Migration Validation

The system shall validate source data before migration:

- Mandatory field completeness
- Data type validation
- Format validation

The system shall identify and flag invalid or incomplete records.

c) Data Quality Checks

The system shall perform data quality checks:

- Duplicate detection
- Referential integrity validation
- Business rule validation

The system shall generate validation reports highlighting issues.

d) Post-Migration Validation

The system shall validate migrated data against source data:

- Record count comparison
- Field-level accuracy checks

The system shall ensure no data loss during migration.

e) Error Handling

- The system shall log all validation errors with detailed descriptions.
- The system shall allow correction and reprocessing of failed records.
- The system shall maintain audit logs of validation activities.

f) Business Validation Support

- The system shall provide reports and tools for business users to validate migrated data.
- The system shall support sampling and detailed verification processes.

10.7 Migration Acceptance

a) Acceptance Criteria Definition

The system shall define measurable acceptance criteria for migration:

- Data completeness
- Data accuracy threshold (e.g., ≥ 99%)
- Integrity of relationships

b) Reconciliation Reports

The system shall generate reconciliation reports comparing source and target data:

- Record counts
- Critical field values

The system shall highlight discrepancies for resolution.

c) User Acceptance Validation

The system shall allow business stakeholders to validate migrated data through reports and system access. The system shall support:

- Functional validation of stakeholder profiles
- Verification of engagement history continuity

11 Testing and Quality Assurance

The Testing and Quality Assurance section defines the processes, activities, and requirements to ensure that the I.M.O Strategic Engagement Platform meets functional, technical, security, and performance expectations before production deployment. The system shall undergo multiple levels of testing to validate end-to-end functionality, data integrity, security compliance, and business usability.

11.1 Functional Testing

Functional Testing shall be conducted to verify that all system functionalities operate in accordance with the defined requirements and business rules.

The system shall support validation of all modules including stakeholder management, engagement tracking, communication logging, and reporting. Each functionality shall be tested against predefined test cases derived from the Functional Requirements Specification. The system shall ensure correct data processing, accurate calculations, workflow execution, and proper handling of user inputs.

The system shall support detection and logging of defects, with traceability to requirement IDs. All identified defects shall be resolved and re-tested prior to progression to subsequent testing phases. Regression testing shall be performed to ensure that fixes or enhancements do not impact existing functionalities.

11.2 Integration Testing

Integration Testing shall be performed to validate interactions between the I.M.O Strategic Engagement Platform and external or internal integrated systems, if any.

The system shall ensure seamless data exchange between connected systems such as communication tools, analytics systems, and document repositories. Data interfaces shall be tested for accuracy, completeness, and consistency of information flow.

The system shall validate API integrations, batch integrations, and real-time data synchronization mechanisms. Error handling during integration failures shall be tested to ensure proper logging, retry mechanisms, and data consistency. The system shall ensure that integrated workflows function correctly across systems without data loss or duplication.

11.3 Security Testing

Security Testing shall be conducted to ensure that the system complies with organizational security standards and safeguards sensitive stakeholder information.

The system shall enforce role-based access control, ensuring that users can only access data relevant to their roles and responsibilities. Authentication and authorization mechanisms shall be tested for robustness.

The system shall be tested for vulnerabilities including unauthorized access, data breaches, injection attacks, and improper data exposure. Data encryption shall be validated for data in transit and at rest where applicable.

The system shall ensure compliance with applicable data protection and privacy regulations. Security audit logs shall be verified to ensure traceability of system access and user activities.

11.4 Performance Testing

Performance testing shall verify that the I.M.O Strategic Engagement Platform performs efficiently under expected and peak workloads while maintaining acceptable response times.

Performance testing shall include:

- System response time testing.
- Load testing with concurrent users.
- Stress testing under peak transaction volumes.
- Scalability testing.
- Database performance testing.
- Report generation performance.
- File upload and download performance.
- Resource utilization monitoring for CPU, memory, storage, and network.

The platform shall maintain stable performance without service interruption under defined operational workloads.

11.5 User Acceptance Testing (UAT)

User Acceptance Testing shall be conducted to validate that the system meets business requirements and is fit for operational use.

Business users shall execute predefined UAT scenarios covering real-world use cases such as stakeholder onboarding, engagement recording, and reporting. The system shall support validation of business workflows, usability, and data accuracy.

The system shall provide UAT environments with representative data for testing. All issues identified during UAT shall be documented, tracked, and resolved prior to final acceptance.

Formal sign-off shall be obtained from business stakeholders confirming that the system meets expected functional and operational requirements.

11.6 Production Readiness Testing

Production Readiness Testing shall be conducted before the system is deployed to the live production environment to verify operational readiness.

Production readiness activities shall include:

- Verification of production environment configuration.

- Database deployment validation.
- Application deployment verification.
- Backup and recovery testing.
- Disaster recovery verification.
- Security configuration validation.
- Performance verification in the production environment.
- Integration verification with production services.
- Monitoring and logging verification.
- Final smoke testing of all critical business functions.

Production deployment shall proceed only after all critical defects have been resolved, testing has been successfully completed, and formal deployment approval has been obtained from the project stakeholders.

12 Deployment and Go-Live Strategy

The Deployment and Go-Live Strategy defines the approach and requirements for deploying the I.M.O Strategic Engagement Platform into the production environment, ensuring minimal disruption to business operations and a smooth transition for users. The system shall support a structured rollout with adequate preparation, validation, and post-deployment support.

12.1 Environment Setup

The system shall support the setup and configuration of multiple environments including development, testing, staging, and production. Each environment shall be configured to meet the necessary infrastructure, security, and performance requirements.

The system shall ensure that all application components, integrations, and dependencies are properly installed and configured in each environment. Environment-specific configurations such as access controls, database connections, and API endpoints shall be validated prior to deployment.

The system shall support version control and deployment management to ensure consistency across environments. Backup and recovery configurations shall be established and verified before production deployment.

12.2 Pilot Rollout

The system shall support a pilot rollout to a limited group of users or business units prior to full-scale deployment. The pilot rollout shall be used to validate system functionality, usability, and performance in a real-world environment.

The system allow monitoring of pilot usage, feedback collection, and issue tracking. Any defects or gaps identified during the pilot phase shall be addressed before proceeding to full deployment.

The system shall ensure that pilot users represent key business roles to provide comprehensive feedback on workflows, engagement tracking, and reporting capabilities.

12.3 User Training

The system shall support structured user training to ensure successful adoption of the platform. Training

should be provided to different user groups based on their roles and responsibilities within the I.M.O Strategic Engagement Platform.

The system shall provide training materials including user guides, process documentation, and reference manuals. The system shall support training sessions such as instructor-led sessions, workshops, or self-paced learning modules.

The system shall ensure that users are trained in key functionalities such as stakeholder management, engagement recording, reporting, and system navigation. Training effectiveness should be evaluated through feedback and assessments where applicable.

12.4 Go-Live Support

The system shall provide dedicated support during the go-live phase to address any issues encountered by users. The system shall ensure availability of technical and functional support teams to manage incidents and user queries.

The system shall support real-time monitoring of system performance and usage during go-live. Any critical issues shall be prioritized and resolved promptly to minimize disruption.

The system shall provide communication channels for users to report issues and seek assistance. Known issues and resolutions shall be documented and shared with users as needed.

12.5 Hypercare Support

The system shall provide hyper care support for a defined period following go-live to ensure system stability and user confidence. During this phase, the system shall support enhanced monitoring, rapid issue resolution, and continuous improvement.

The system should track and prioritize issues reported during hyper care, with faster turnaround times for resolution. The system shall support regular status reporting on system health, issue trends, and resolution progress.

The system shall ensure knowledge transfer to the steady-state support team at the end of the hyper care period. The system shall gradually transition from hyper care to standard support operations once stability and performance criteria are met.

13 Acceptance Criteria

The Acceptance Criteria section defines the measurable conditions that must be met for the I.M.O Strategic Engagement Platform to be considered complete, validated, and ready for production use. These criteria ensure that the system meets business, functional, performance, and security expectations prior to final approval.

13.1 Functional Acceptance Criteria

The system shall meet all functional requirements as defined in the Functional Requirements Specification. All modules, including stakeholder management, engagement tracking, communication logging, reporting, and workflows, shall operate accurately and consistently.

The system shall ensure that all business processes are executed as intended without errors or data inconsistencies. All critical and high-priority defects identified during testing phases shall be resolved before acceptance.

The system shall demonstrate complete traceability between requirements, test cases, and test results. The system shall ensure that no critical functionality is missing or partially implemented.

13.2 Performance Acceptance Criteria

The system shall meet defined performance benchmarks for response time, system availability, and data processing.

The system shall ensure acceptable response times for key operations such as stakeholder search, engagement recording, and report generation under normal and peak load conditions.

The system shall support concurrent users without significant degradation in performance. The system shall demonstrate stability during load and stress testing scenarios.

The system shall meet agreed service level expectations for uptime and system reliability.

13.3 Security Acceptance Criteria

The system shall comply with all defined security requirements and organizational policies.

The system shall enforce role-based access control ensuring that users can only access authorized data and functionalities. Authentication and authorization mechanisms shall function correctly without vulnerabilities.

The system shall ensure protection of sensitive stakeholder and engagement data through encryption and secure data handling practices.

All identified security vulnerabilities shall be resolved, and the system shall pass required security assessments such as vulnerability scans and penetration testing.

Audit logging shall be enabled and validated to ensure traceability of user activities.

13.4 UAT Sign-Off Criteria

The system shall successfully complete User Acceptance Testing with validation from business stakeholders.

All UAT scenarios shall be executed with expected results meeting business requirements. Any defects identified during UAT shall be resolved or formally accepted with documented justification.

The system shall demonstrate usability and alignment with real-world business processes. Business users shall confirm that the system supports their operational needs effectively.

Formal sign-off shall be obtained from designated business representatives confirming acceptance of the system for production use.

13.5 Go-Live Approval Criteria

The system shall meet all readiness conditions prior to go-live, including successful completion of testing phases, migration validation, and environment setup.

All critical defects shall be resolved, and no open issues shall pose a risk to production operations. Required documentation, including user guides and support procedures, shall be completed and available.

The system shall demonstrate readiness in terms of infrastructure, integrations, security configurations, and monitoring capabilities.

Final approval for go-live shall be obtained from both business and technical stakeholders, confirming that the system is stable, compliant, and ready for deployment into the production environment.

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